

1 Mark Joseph Nejme
2 P.O. Box 589
3 Clifton, New Jersey 07012
4 Telephone: 732-995-3914
5 Email: research@warrantwire.com
6 Plaintiff, self-represented
7

8 UNITED STATES DISTRICT COURT
9 DISTRICT OF NEVADA
10

11 MARK JOSEPH NEJMEH,
12 Plaintiff,

13
14 v.

15
16 THERIVA BIOLOGICS, INC.,
17 Defendant.
18
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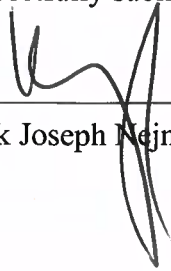
Case No. 3:26-cv-00705

PLAINTIFF'S MOTION FOR LEAVE
TO FILE SUPPLEMENTAL
SUMMARY IN SUPPORT OF
INTERIM RELIEF, WITH EXHIBITS
ST-1 AND ST-2

20 Plaintiff respectfully moves under Local Rule 7-2(g) for leave to file the attached
21 Supplemental Summary in Support of Interim Relief, with Exhibits ST-1 and ST-2. Good
22 cause exists. The Summary adds no claim or party. It draws together, from Defendant's own
23 SEC filings, the anti-takeover powers on which the corporate actions identified in Plaintiff's
24 Supplemental Statement depend. Those passages appear deep inside a 36-page prospectus
25 supplement and a 50-page financing agreement; the attached exhibits reproduce the
26 complete filings with the cited passages highlighted and the page of each identified, so the
27 Court can verify them directly. The Summary bears on the pending Emergency Motion
28 (ECF No. 2) and on the interim relief already requested, and Defendant may respond to it
29 within the time the Court sets.

30 Dated: September 29, 2026

31 Respectfully submitted,

32 
33 _____
34 Mark Joseph Nejme, Plaintiff, self-represented
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16 THERIVA BIOLOGICS, INC.,
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Case No. 3:26-cv-00705

PLAINTIFF'S SUPPLEMENTAL
SUMMARY IN SUPPORT OF
INTERIM RELIEF: THE PLAN
AGAINST DEFENDANT'S
STOCKHOLDERS AND THE ANTI-
TAKEOVER POWERS IT DEPENDS
ON

21 1. Plaintiff submits this summary in support of his Emergency Motion for a
22 Temporary Restraining Order and/or Preliminary Injunction (ECF No. 2) and his
23 Supplemental Statement in Support of Interim Relief, subject to the Court granting leave
24 under Local Rule 7-2(g). It adds no claim or party. Part I explains the matter in plain
25 English. Part II sets out what Defendant's own filings say. Part III states, step by step, the
26 plan Plaintiff contends Defendant is using against its stockholders, with the record source
27 for each step. Part IV addresses the directors' duties and Part V the relief. Part VI is
28 Plaintiff's closing argument: that this has been a plan since 2021. Part VII lists the exhibits.
29

30 I. IN PLAIN ENGLISH

31 1A. The company has a weapon that keeps buyers away. Theriva's board can
32 create up to ten million shares of a special class of stock, called preferred stock, whenever it
33 wants, without asking the stockholders. Theriva itself says in its filings that this stock
34 "could operate as a 'poison pill'" to stop anyone from buying the company unless the board
35 approves. A poison pill works by flooding the market with new shares so that any outside
36 buyer's stake is watered down until buying the company is no longer worth it.

1 **1B. The board promised its financiers that the weapon will never be used on**
2 **them.** In May 2025, Theriva signed a written promise to the funds that buy its discounted
3 shares and warrants: no poison pill, whether it already exists or is adopted later, will ever be
4 counted against those funds because of the shares and warrants they get from Theriva. No
5 outside buyer has that promise. Defendant may call this standard contract language.
6 Plaintiff contends it is a special favor, and the pattern shows it: the same small group of
7 funds comes back deal after deal, each time buying new stock at a discount, while ordinary
8 stockholders are diluted.

9 **1C. So only one side is protected.** The people who keep buying cheap new stock
10 from the company are shielded. Anyone who might offer the stockholders a fair price for
11 the whole company, and for its two valuable drug programs, VCN-01 and SYN-004, faces
12 the poison pill. That shuts outsiders out and removes the fair-market opportunity that
13 stockholders would otherwise have: a buyer bidding for what the company is really worth.

14 **1D. The stockholders were never told this in plain English.** These facts are not
15 explained to stockholders in plain words. They are buried on page 19 of a 36-page
16 prospectus and in two clauses of a 50-page financing contract, surrounded by pages of legal
17 and technical language. Plaintiff has attached both documents in full, with the passages
18 highlighted, so the Court can see exactly where they are.

19 **1E. The stockholders have been substantially damaged.** The company keeps
20 issuing new shares and warrants to the same funds, has reverse-split its stock, and in August
21 2026 obtained approval for 16,184,560 more warrant shares, equal to about 35% of all
22 shares now outstanding. Each round leaves every existing stockholder owning a smaller
23 piece of the company. Meanwhile the company says it is considering a reverse merger, in
24 which a private company typically takes over the public company and its drug programs
25 and the existing stockholders are left with a small fraction. The poison pill is what keeps a
26 better offer from coming in while this happens.

27 **1F. The plan, in plain English.** Plaintiff contends the plan works like this:

28 (1) Sell new shares and warrants, at a discount, to the same small group of funds,
29 again and again. Every sale gives each existing stockholder a smaller piece of the
30 company.

31 (2) When the share price falls too low, do a reverse split: turn many shares into one,
32 without asking the stockholders. The price looks higher; nobody's piece gets bigger.

33 (3) Write the stock-plan and authorized-share numbers so that a reverse split does
34 not shrink them. After the next split, the board can issue far more new stock,
35 compared with what the stockholders hold, than it could before.
36

1 (4) Keep the poison pill ready so that no outside buyer can come in and offer the
2 stockholders what the company and its drugs are really worth. Promise the funds, in
3 writing, that the pill will never be used against them.

4 (5) Announce a “strategic review” that may end in a reverse merger, and give the
5 chief executive change-in-control pay if it does.

6 (6) Let the drugs go: one has already been licensed away for \$300,000 up front,
7 another is no longer funded, and the most valuable one, VCN-01, is what a reverse-
8 merger partner would receive.

9 (7) When it is done, the stockholders who owned the company and its drugs are left
10 with a small piece of something else.

11 **1G. What a poison pill is, in plain English.** Imagine you own a share of a house
12 with other owners, and a buyer offers a fair price for the whole house. Now imagine the
13 manager of the house can, whenever he wants and without asking the owners, print
14 thousands of new ownership papers and hand them out, so that the buyer’s offer suddenly
15 buys only a sliver of the house. No buyer will make an offer under those conditions. That is
16 a poison pill: it does not stop a sale by saying no; it stops a sale by making sure no buyer
17 ever shows up. The owners never hear the offer they might have received. At Theriva, the
18 “new ownership papers” are the ten million shares of preferred stock the board may issue
19 on its own (Ex. ST-1 at 19). The danger is not theoretical: in July 2022 Defendant sold a
20 single investor Series D preferred stock carrying 20,000 votes per share in the stockholder
21 vote on increasing its authorized common stock to 350,000,000 shares. (Supplemental
22 Exhibit SH-2; Supplemental Exhibit SS ¶ 52.) The board has already used preferred stock to
23 decide a stockholder vote.

24 **1H. What is at stake, in dollars.** Supplemental Exhibit SL puts numbers on it.
25 Revolution Medicines (RVMD) is a pancreatic-cancer drug company. In its Phase 3 trial,
26 patients on its drug lived a median 13.2 months. In Theriva’s trial, patients who received
27 two doses of VCN-01 lived a median 14.8 months, 1.6 months longer. On September 22,
28 2026, the stock market valued RVMD at about \$41.34 billion and Theriva at about \$10.59
29 million, roughly 3,900 times less. Spread across Theriva’s 45,892,668 shares, a company
30 valued like RVMD would be worth about \$900.80 per share. Theriva’s stock closed that day
31 at about 23 cents. (Ex. SL-1, SL-2.) That is before counting Theriva’s second drug, SYN-
32 004, which in a 412-patient controlled trial cut *C. difficile* infections by 71.4%. *C. difficile*
33 causes almost 500,000 infections in the United States each year, at an estimated annual cost
34 of about \$5.4 billion to \$6.3 billion. Plaintiff’s scenario of \$2.0 billion for SYN-004 adds
35 about \$43.58 per share. (Ex. SL-2, SL-3.) Together, that is about \$944 per share, or about
36 \$43 billion, for a company the market prices at about \$10.6 million. Plaintiff does not ask

1 the Court to find that Theriva is worth that amount; the trials differ, and Exhibit SL itself
2 says the comparison is one of scale, not an appraisal. Plaintiff contends that value on that
3 scale is what the plan is built to capture, and it is exactly the kind of value a poison pill
4 keeps outside buyers from bidding for.

5 6 **II. WHAT DEFENDANT'S OWN FILINGS SAY**

7 **2. Ten million shares of preferred stock that can act as a "poison pill."**

8 Defendant's prospectus supplement filed December 3, 2025 (Form 424B5, accession
9 0001104659-25-118116; Exhibit ST-1) states that its authorized capital includes
10 "10,000,000 shares of preferred stock" (Ex. ST-1 at 15), and that "our board of directors
11 will be authorized to issue, without stockholder approval, preferred stock, the rights of
12 which will be determined at the discretion of the board of directors and that, if issued, could
13 operate as a 'poison pill' to dilute the stock ownership of a potential hostile acquirer to
14 prevent an acquisition that our board of directors does not approve." (Ex. ST-1 at 19.)

15 **3. Shares for "persons friendly to current management."** On the same page
16 Defendant states: "The existence of unissued and unreserved Common Stock may enable
17 our board of directors to issue shares to persons friendly to current management." (Ex. ST-1
18 at 19.)

19 **4. The Nevada business-combination statute.** Under the heading "Anti-Takeover
20 Effects of Nevada Law," Defendant states that these provisions "may have the effect of
21 delaying, deferring or discouraging another party from acquiring control of us," and that
22 Nev. Rev. Stat. §§ 78.411–78.444 "may discourage attempts to acquire our company." (Ex.
23 ST-1 at 15, 17.)

24 **5. The financiers were promised protection from the poison pill.** In the Securities
25 Purchase Agreement filed with Defendant's Form 8-K of May 8, 2025 (Exhibit 10.1,
26 accession 0001104659-25-046170; Exhibit ST-2), Defendant represented that it and its
27 board "have taken all necessary action, if any, in order to render inapplicable any control
28 share acquisition, business combination, poison pill (including any distribution under a
29 rights agreement) or other similar anti-takeover provision" that could apply to the
30 purchasers (§ 3.1(x); Ex. ST-2 at 24), and agreed that "[n]o claim will be made or enforced
31 by the Company" that any purchaser is an "Acquiring Person" under any such plan or
32 arrangement "in effect or hereafter adopted by the Company" by virtue of receiving
33 securities from Defendant (§ 4.5, "Stockholder Rights Plan"; Ex. ST-2 at 34). No outside
34 buyer holds any such promise.

Step 6: Prepare a reverse merger. Defendant’s August 2026 Form 10-Q states that it is exploring “an acquisition, merger, reverse merger, other business combination, sales of assets or other strategic transactions” and is “devoting substantial time and resources to the strategic review.” The chief executive’s amended employment agreement of August 10, 2026 pays change-in-control benefits on a reverse merger with a non-listed private company, and a “strategic transaction” makes the 6,500,000-share plan limit effective before January 1, 2027. (Supplemental Exhibit SN-4; Supplemental Statement ¶¶ 5(a), 7.)

Step 7: Move the drugs. Defendant has already licensed SYN-020 exclusively and worldwide for \$300,000 up front and has stated that it will not fund further SYN-004 development internally. VCN-01 holds FDA Orphan Drug, Fast Track and Rare Pediatric Disease designations and FDA's general agreement with its proposed Phase 3 design. (Supplemental Exhibit SR ¶ 19; Supplemental Statement ¶ 8.) Those programs are what a reverse-merger partner or transferee would obtain.

Step 8: The same directors have done it before. Two of Defendant's directors, who sit on all three of its board committees, serve or served on the board of Heat Biologics, which sold its subsidiary owning an FDA-approved product to a company controlled by its own chairman, later renegotiated its terms, and lost its remaining assets to foreclosure after delisting. (Supplemental Exhibit SR ¶¶ 4–18; Supplemental Statement ¶ 9.) Plaintiff does not contend those transactions were unlawful; he cites them because they show how such transfers can be completed before public stockholders have any practical opportunity to be heard.

7. The result. Plaintiff contends that, taken together, these steps let Defendant's board dilute the existing stockholders, reverse-split them without a vote while its issuing power stays the same, keep any buyer willing to pay fair value out of the room, and move VCN-01 and SYN-004 into a reverse merger or transfer on terms set by insiders and the protected financiers. The plan works only because the anti-takeover powers in Part II stop any outside party from competing. In Plaintiff's view, its effect is to take two valuable drug programs from the stockholders who own them and to destroy the value of their shares.

IV. THE DIRECTORS' DUTIES

8. Under Nevada law, “[t]he fiduciary duties of directors and officers are to exercise their respective powers in good faith, on an informed basis and with a view to the interests of the corporation.” Nev. Rev. Stat. § 78.138(1). Plaintiff recognizes that Nevada gives boards wide room to adopt rights plans, Nev. Rev. Stat. § 78.139(2)(b), but that room does not “permit directors or officers to abrogate any right conferred by the laws of this State or

1 the articles of incorporation,” § 78.139(3), and it does not displace the duty of good faith in
2 § 78.138(1). Plaintiff contends that using, or keeping ready, the power to issue preferred
3 stock as a “poison pill” against outside buyers, while shielding the financiers, in order to
4 carry out the plan described in Part III, would not be an exercise of the board’s powers in
5 good faith and with a view to the interests of the corporation and its stockholders, and
6 would breach the fiduciary duties the directors owe. It should not be allowed.

7 8 **V. RELIEF**

9 **9.** Consistent with paragraphs 2(b) through 2(d) of the Supplemental Statement,
10 Plaintiff respectfully asks that, pending a noticed hearing on ECF No. 2, the Court order
11 that Defendant (a) not issue any shares of preferred stock, and not adopt, amend or trigger
12 any shareholder-rights plan (“poison pill”) or similar instrument; (b) not approve or effect a
13 reverse stock split or a reverse merger; or, in the alternative to (a) and (b), file and serve
14 notice at least ten business days before doing any of those things, stating its terms; and (c)
15 file and serve notice at least ten business days before signing any agreement for a merger,
16 reverse merger, business combination, change of control, or transfer of all or substantially
17 all of the rights to VCN-01 or SYN-004, so that stockholders and the Court may address an
18 actual agreement before it takes effect.

19 20 **VI. PLAINTIFF’S CLOSING ARGUMENT: A PLAN SINCE 2021**

21 **10.** Plaintiff contends that none of this is an accident, and none of it is new. Taken
22 year by year, Defendant’s own filings show the same plan carried out step by step since at
23 least 2021:

24 **2021.** On January 27, 2021, Defendant cut the conversion price of MSD Credit
25 Opportunity Master Fund’s Series A preferred stock from \$18.90 to \$1.50 per share
26 and removed MSD’s right to be repurchased on a change of control; MSD converted
27 into 8,996,768 common shares in January and February 2021, and Defendant
28 recorded a \$7.402 million deemed dividend for the inducement. (Supplemental
29 Exhibit SS ¶ 52; Supplemental Exhibit SA-5.) The same year, Hudson Bay Capital
30 Management, a Defendant financier since 2011, reported holding warrants on
31 2,219,019 shares at a 9.99% cap; it now holds 3,363,640 of the New Warrants.
32 (Supplemental Exhibit SS ¶ 48.)

33 **2022.** Defendant sold MSD Series C and Series D preferred stock, the Series D
34 carrying 20,000 votes per share in the vote on increasing authorized common stock
35 to 350,000,000 shares, and reverse-split its stock 1-for-10. (Supplemental Exhibit
36 SH-2; Supplemental Exhibit SS ¶ 52; Supplemental Exhibit SQ.)

2024. Defendant's board reverse-split the stock again, 1-for-25, without a stockholder vote. (Supplemental Statement ¶ 4.)

2025. Defendant filed nine offering prospectuses (Supplemental Exhibit SQ), gave its financiers the written promise that no poison pill would be used against them on account of the stock they receive (Ex. ST-2 at 24, 34), and told the market that its board may issue preferred stock as a “poison pill” against any acquisition the board does not approve (Ex. ST-1 at 19).

2026. Defendant obtained approval of 16,184,560 more warrant shares, fixed its stock-plan and authorized-share numbers so a reverse split will not shrink them, announced a strategic review that may include a reverse merger, and amended its chief executive’s agreement to pay change-in-control benefits on a reverse merger with a non-listed private company. (Supplemental Statement ¶¶ 5–8; Supplemental Exhibit SN-4.)

11. Over those years, on a split-adjusted basis, Defendant's share count grew from about 0.5 million to about 45.9 million shares. (Supplemental Exhibits SA-SP, share-count chart.) Every step pointed the same way: more stock for the same favored parties, less of the company for everyone else, and a locked door against any buyer who might pay the stockholders what their company is worth. Plaintiff contends that a pattern this consistent, over this many years, by the same directors, is not a series of unrelated business decisions. It was premeditated. The plan has five moves: first, the reverse splits; second, the warrants; third, the preparation of the poison pill; fourth, the reverse merger set up in advance; and last, the poison pill itself, which keeps every outside buyer away while the technology, VCN-01 and SYN-004, is taken from the stockholders who own it. On the scale shown in Exhibit SL, that technology stands for value of roughly \$900 per share, about \$41 billion, plus the C. difficile program, while the stockholders are left holding shares the market prices at about 23 cents. (Ex. SL-1 to SL-3; paragraph 1H.) Plaintiff asks the Court to stop it before the last move is made.

VII. EXHIBITS AND LIMITS

12. Exhibit ST-1 is Defendant's complete prospectus supplement filed December 3, 2025, and Exhibit ST-2 is the complete Securities Purchase Agreement filed as Exhibit 10.1 to Defendant's Form 8-K of May 8, 2025, each as published on the SEC's EDGAR system, with highlighting added by Plaintiff to mark the cited passages. The Court may take judicial notice of these public filings. Exhibit ST-2 is a contract between Defendant and its purchasers; its terms, including its waiver of jury trial (§ 5.21; Ex. ST-2 at 45), bind only the parties to it (§ 5.8, "No Third-Party Beneficiaries") and do not apply to Plaintiff, who is

1 not a party to it and has demanded trial by jury (ECF No. 7). Plaintiff offers Exhibit ST-2
2 only as evidence of what Defendant agreed to, and waives nothing by filing it. Paragraphs
3 1F, 1G, 1H, 6 through 8, 10 and 11 state Plaintiff's contention drawn from the record; the
4 interim relief requested preserves the present position and does not require the Court to
5 adopt that contention. Plaintiff reserves the right to seek leave to amend. He understands
6 that any restraint must satisfy Federal Rule of Civil Procedure 65, including specificity
7 under Rule 65(d) and any security the Court finds proper under Rule 65(c).

8 Dated: September 29, 2026

9 Respectfully submitted,

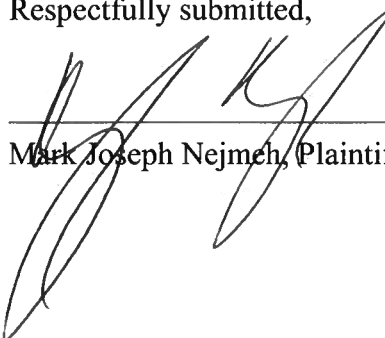
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EXHIBIT ST-1

Nejmeh v. Theriva Biologics, Inc., Case No. 3:26-cv-00705 (D. Nev.)

Theriva Biologics, Inc., Prospectus Supplement (Form 424B5), filed December 3, 2025

SEC accession no. 0001104659-25-118116

https://www.sec.gov/Archives/edgar/data/894158/000110465925118116/tm2532622-1_424b5.htm

Cited passages: authorized capital of 10,000,000 shares of preferred stock and “Anti-Takeover Effects of Nevada Law” (Exhibit pages 15 and 17; prospectus pages 7–8); the board’s power to issue preferred stock without stockholder approval that “could operate as a ‘poison pill’,” and unissued stock that may be issued “to persons friendly to current management” (Exhibit page 19; prospectus page 9).

Yellow highlighting has been added by Plaintiff to mark the passages cited in Plaintiff’s Supplemental Summary. The text of the filing is otherwise reproduced as published on the SEC’s EDGAR system.

**16,184,560 Shares
Of Common Stock**

This prospectus relates to the resale from time to time of up to an aggregate of 16,184,560 shares of common stock, par value \$0.001 per share (the “Common Stock”), of Theriva Biologics, Inc. by the Selling Stockholders identified in this prospectus (the “Selling Stockholders”), including their pledgees, assignees, donees, transferees or their respective successors-in-interest. The 16,184,560 shares of Common Stock registered hereunder are issuable upon exercise of New Warrants (defined below) that were issued to the Selling Stockholders on October 16, 2025 when we entered into a warrant inducement agreement (the “Warrant Inducement Agreement”) with the Selling Stockholders, which were holders of previously issued common stock purchase warrants to purchase up to an aggregate of 8,092,280 shares of Common Stock, consisting of (i) common stock purchase warrants to purchase up to an aggregate of 1,345,000 shares of Common Stock issued on September 27, 2024 (the “September 2024 Warrants”, and (ii) common stock purchase warrants to purchase up to an aggregate of 6,747,280 shares of Common Stock issued on May 8, 2025 (the “May 2025 Warrants” and, together with the September Warrants, the “Existing Warrants”). Pursuant to the Warrant Inducement Agreement, (1) the holders of the Existing Warrants (who are the Selling Stockholders named herein) agreed to exercise for cash the Existing Warrants at a reduced exercise price of \$0.54 per share and (2) we agreed to issue to the Selling Stockholders new common stock purchase warrants (the “New Warrants”) to purchase up to an aggregate of 16,184,560 shares of Common Stock (sometimes referred to as the “New Warrant Shares”).

We are filing this registration statement on Form S-3, of which this prospectus forms a part, to fulfill our contractual obligations with the Selling Stockholders to provide for the resale by the Selling Stockholders of the 16,184,560 shares of Common Stock offered hereby. See “Selling Stockholders” beginning on page 13 of this prospectus for more information about the Selling Stockholders. The registration of the shares of Common Stock to which this prospectus relates does not require the Selling Stockholders to sell any of their shares of Common Stock. We are not offering any shares of Common Stock under this prospectus and will not receive any proceeds from the sale or other disposition of the shares of Common Stock covered hereby. See “Use of Proceeds” beginning on page 6 of this prospectus.

The Selling Stockholders identified in this prospectus, or their respective pledgees, assignees, donees, transferees or their respective successors-in-interest, from time to time may offer and sell through public or private transactions at prevailing market prices, at prices related to prevailing market prices or at privately negotiated prices the shares held by them directly or through underwriters, agents or broker-dealers on terms to be determined at the time of sale, as described in more detail in this prospectus. See “Plan of Distribution” beginning on page 16 of this prospectus for more information about how the Selling Stockholders may sell their respective shares of Common Stock. The Selling Stockholders may be deemed “underwriters” within the meaning of Section 2(a)(11) of the Securities Act of 1933, as amended (the “Securities Act”).

Pursuant to the terms of the Warrant Inducement Agreement, we have agreed to bear all of the expenses in connection with the registration of the New Warrant Shares pursuant to the registration statement of which this prospectus forms a part. The Stockholders will pay or assume all commissions, discounts, fees of underwriters, agents, selling brokers or dealer managers and similar expenses, if any, attributable to their respective sales of the New Warrant Shares.

Our shares of Common Stock are listed on the NYSE American LLC (the “NYSE American”) under the symbol “TOVX”. On December 1, 2025, the closing price for our shares of Common Stock on the NYSE American was \$0.2189 per share.

Investing in our Common Stock involves risks. You should review carefully the risks described under the heading “Risk Factors” contained in this prospectus and under similar headings in the other documents that are incorporated by reference into this prospectus, as described beginning on page 4 of this prospectus.

Neither the Securities and Exchange Commission nor any state securities commission has approved or disapproved of these securities or passed upon the adequacy or accuracy of this prospectus. Any representation to the contrary is a criminal offense. The securities are not being offered in any jurisdiction where the offer is not permitted.

The date of this prospectus is December 2, 2025

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ABOUT THIS PROSPECTUS

You should rely only on the information we have provided or incorporated by reference into this prospectus and any related free writing prospectus. We have not authorized anyone to provide you with information different from that contained in this prospectus, any applicable prospectus or any related free writing prospectus. No dealer, salesperson or other person is authorized to give any information or to represent anything not contained in this prospectus, any applicable prospectus or any related free writing prospectus. You must not rely on any unauthorized information or representation. This prospectus is an offer to sell only the shares of Common Stock offered hereby, but only under circumstances and in jurisdictions where it is lawful to do so. You should assume that the information in this prospectus or any related free writing prospectus is accurate only as of the date on the front of the document and that any information we have incorporated by reference is accurate only as of the date of the document incorporated by reference, regardless of the time of delivery of this prospectus or any sale of a security.

This prospectus and the documents incorporated by reference into this prospectus include statistical and other industry and market data that we obtained from industry publications and research, surveys and studies conducted by third parties. Industry publications and third-party research, surveys and studies generally indicate that their information has been obtained from sources believed to be reliable, although they do not guarantee the accuracy or completeness of such information. We believe that the data obtained from these industry publications and third-party research, surveys and studies are reliable. We are ultimately responsible for all disclosure included in this prospectus.

The Selling Stockholders are offering shares of Common Stock only in jurisdictions where such issuances are permitted. The distribution of this prospectus and the issuance of the shares of Common Stock in certain jurisdictions may be restricted by law. Persons outside the United States who come into possession of this prospectus must inform themselves about, and observe any restrictions relating to, the issuance of the shares and the distribution of this prospectus outside the United States. This prospectus does not constitute, and may not be used in connection with, an offer to sell, or a solicitation of an offer to buy, the shares of Common Stock offered by this prospectus by any person in any jurisdiction in which it is unlawful for such person to make such an offer or solicitation.

This prospectus contains summaries of certain provisions contained in some of the documents described herein, but reference is made to the actual documents for complete information. All of the summaries are qualified in their entirety by the actual documents. Copies of some of the documents referred to herein have been filed, will be filed or will be incorporated by reference as exhibits to the registration statement of which this prospectus is a part, and you may obtain copies of those documents as described below under the sections entitled "Where You Can Find More Information" and "Incorporation of Certain Documents By Reference."

Neither we nor any Selling Stockholders has authorized anyone to provide you with information different from that contained in this prospectus, any accompanying prospectus or in any related free-writing prospectus filed by us with the Securities and Exchange Commission (the "SEC"). Neither we nor any Selling Stockholder takes any responsibility for, or provides any assurance as to the reliability of, any information other than the information in this prospectus, any accompanying prospectus or in any related free-writing prospectus filed by us with the SEC. This prospectus and any accompanying prospectus do not constitute an offer to sell or the solicitation of an offer to buy any securities other than the securities described in this prospectus or any accompanying prospectus or an offer to sell or the solicitation of an offer to buy such securities in any circumstances in which such offer or solicitation is unlawful. You should assume that the information appearing in this prospectus, any prospectus, the documents incorporated by reference and any related free-writing prospectus is accurate only as of their respective dates. Our business, financial condition, results of operations and prospects may have changed materially since those dates.

Except as otherwise indicated herein or as the context otherwise requires, references in this prospectus to "Theriva," "the Company," "we," "us," "our" and similar references refer to Theriva Biologics, Inc., an entity incorporated under the laws of the State of Delaware.

Smaller Reporting Company – Scaled Disclosure

Pursuant to Item 10(f) of Regulation S-K promulgated under the Securities Act of 1933, as amended, as indicated herein, we have elected to comply with the scaled disclosure requirements applicable to “smaller reporting companies,” including providing two years of audited financial statements.

PROSPECTUS SUMMARY

This summary highlights information contained in greater detail elsewhere in this prospectus. This summary is not complete and does not contain all of the information you should consider in making your investment decision. You should read the entire prospectus carefully before making an investment in our securities. You should carefully consider, among other things, our financial statements and the related notes and the sections entitled “Risk Factors” and “Management’s Discussion and Analysis of Financial Condition and Results of Operations” included elsewhere in, or incorporated by reference into, this prospectus.

The Company

Overview

We are a diversified clinical-stage company developing therapeutics designed to treat cancer and related diseases in areas of high unmet need. As a result of the acquisition in March 2022 of Theriva Biologics, S.L. (“VCN”, formerly named VCN Biosciences, S.L.), described in more detail below (the “Acquisition”), we transitioned our strategic focus to oncology through the development of VCN’s new oncolytic adenovirus platform designed for intravenous and intravitreal delivery to trigger tumor cell death, to improve access of co-administered cancer therapies to the tumor, and to promote a robust and sustained anti-tumor response by the patient’s immune system. Our lead product candidate, VCN-01, a clinical stage oncolytic human adenovirus that is modified for tumor-selective replication and to express an enzyme, PH20 hyaluronidase, has been evaluated in a Phase 2b clinical study for the treatment of pancreatic cancer (“VIRAGE”), and has recently been used to treat patients in a Phase 1 clinical study for the treatment of retinoblastoma, and Phase 1 clinical studies for the treatment of other solid tumors including head and neck squamous cell carcinoma.

Prior to the Acquisition, our focus was on developing therapeutics designed to treat gastrointestinal (GI) diseases which included our clinical development candidates: (1) SYN-004 (ribaxamase) which is designed to degrade certain commonly used intravenous (IV) beta-lactam antibiotics within the GI tract to prevent microbiome damage, thereby preventing overgrowth and infection by pathogenic organisms such as *Clostridioides difficile* infection (CDI) and vancomycin resistant Enterococci (VRE), and reducing the incidence and severity of acute graft-versus-host-disease (aGVHD) in allogeneic hematopoietic cell transplant (HCT) recipients, and (2) SYN-020, a recombinant oral formulation of the enzyme intestinal alkaline phosphatase (IAP) produced under cGMP conditions and intended to treat both local GI and systemic diseases. As part of our strategic transformation into an oncology focused company, we are exploring value creation options for our SYN-004 and SYN-020 assets, including out-licensing or partnering.

Our Current Product Pipeline

****Based on management's current beliefs and expectations***

aGVHD acute graft-versus host disease. **allo-HCT** allogeneic hematopoietic cell transplant. **CSR** clinical study report. **HNSCC** head and neck squamous cell carcinoma. **IV** intravenous. **IVit** intravitreal. For other abbreviations see the text.

¹Additional products with preclinical proof-of-concept include SYN-006 (carbapenemase) to prevent aGVHD, CDI, and microbiome damage in patients treated with carbapenem antibiotics and SYN-007 (ribaxamase) DR to prevent antibiotic associated diarrhea with oral β -lactam antibiotics.

²Depending on funding/partnership. SYN-004 may enter a U.S. Food and Drug Administration ("FDA")-agreed Phase 3 clinical trial for the treatment of CDI.

Corporate Information

Our predecessor, Sheffield Pharmaceuticals, Inc., was incorporated in 1986, and in 2006 engaged in a reverse merger with Pipex Therapeutics, Inc., a publicly-traded Delaware corporation formed in 2001. After the reverse merger, we changed our name to Pipex Pharmaceuticals, Inc., and in October 2008 we changed our name to Adeona Pharmaceuticals, Inc. On October 15, 2009, we engaged in a merger with a wholly owned subsidiary for the purpose of reincorporating in the State of Nevada. On February 15, 2012, we changed our name to Synthetic Biologics, Inc. On August 10, 2018, we effected a one for thirty-five reverse stock split of our authorized, issued and outstanding Common Stock. On July 15, 2022, we effected a one for ten reverse stock split of our authorized, issued and outstanding Common Stock. On October 12, 2022, we changed our name to Theriva Biologics, Inc. On August 26, 2024, we effected a one for twenty-five reverse stock split of our authorized, issued and outstanding Common stock (the "Reverse Stock Split").

Our principal executive offices are located at 9605 Medical Center Drive, Suite 270, Rockville, Maryland 20850, and our telephone number is (301) 417-4367. Our website address is www.therivabio.com. Information contained on our website is intended for informational purposes only and is not incorporated by reference into this prospectus, and it should not be considered to be part of this prospectus or the registration statement of which this prospectus forms a part. The SEC maintains an internet site that contains reports, proxy and information statements, and other information regarding issuers like us that file documents electronically with the SEC. The address of the SEC website is www.sec.gov.

Smaller Reporting Company

We are a "smaller reporting company" as defined in the Securities Exchange Act of 1934, as amended (the "Exchange Act"). As a result, we may take advantage of certain reduced disclosure obligations available to smaller reporting companies, including the exemption from compliance with the auditor attestation requirements pursuant to the Sarbanes-Oxley Act of 2002, reduced disclosure about our executive compensation arrangements and the requirements to provide only two years of audited financial statements in our annual reports and registration statements. We will continue to be a "smaller reporting company" as long as (1) we have a public float (i.e., the market value of our Common Stock held by non-affiliates) less than \$250 million calculated as of the last business day of our most recently completed second fiscal quarter, or (2) our annual revenues are less than \$100 million for our previous fiscal year and we have either no public float or a public float of less than \$700 million as of the end of that fiscal year's second fiscal quarter. Decreased disclosures in our SEC filings due to our status as a "smaller reporting company" may make it harder for investors to analyze our results of operations and financial prospects.

THE OFFERING

Shares of common stock offered by the Selling Stockholders	16,184,560 shares
Common stock to be outstanding after this offering⁽¹⁾	49,924,203 shares of Common Stock, assuming the exercise of all of the New Warrants
Registration Rights	Under the terms of the Warrant Inducement Agreement, we agreed to file this registration statement with respect to the registration of the resale by the Selling Stockholders of the New Warrant Shares as soon as reasonably practicable and in any event by the 30th calendar day following the date of the Warrant Inducement Agreement, and to use commercially reasonable efforts to cause this registration statement of which this prospectus forms a part to become effective within sixty (60) calendar days following the date of the Warrant Inducement Agreement (or within 90 calendar days following the date of the Warrant Inducement Agreement in case of a review of this registration statement by the SEC), provided, however, that if the SEC is closed for operations due to a government shutdown, the applicable deadline for this registration statement to become effective will be extended by the same amount of days that the SEC remains closed for operations, and to keep this registration statement effective at all times until no holder of the New Warrants owns any New Warrants or New Warrant Shares.
Use of Proceeds	The Stockholders will receive all of the proceeds of the sale of shares of Common Stock offered from time to time pursuant to this prospectus. Accordingly, we will not receive any proceeds from the sale of shares of Common Stock that may be sold from time to time pursuant to this prospectus; however, we will receive proceeds from the cash exercise of the New Warrants. See “Use of Proceeds.” We intend to use the proceeds from any cash exercise of the New Warrants for working capital and other general corporate purposes.
Plan of Distribution	The Selling Stockholders named in this prospectus, or their pledgees, donees, transferees, distributees, beneficiaries or other successors-in-interest, may offer or sell the shares of Common Stock offered hereby from time to time through public or private transactions at prevailing market prices, at prices related to prevailing market prices or at privately negotiated prices. The Stockholders may also resell the shares of Common Stock to or through underwriters, broker-dealers or agents, who may receive compensation in the form of discounts, concessions or commissions.
Risk Factors	See “Risk Factors” beginning on page 4 of this prospectus and in the documents incorporated by reference in this prospectus and the other information included in this prospectus for a discussion of factors you should carefully consider before investing in our securities.
NYSE American trading symbol	Our shares of Common Stock are listed on the NYSE American under the symbol “TOVX”.

(1) Based on 33,739,643 shares of Common Stock outstanding as of November 3, 2025, and excludes:

- 1,116,101 shares of Common Stock issuable upon the exercise of outstanding stock options with a weighted average exercise price of \$5.09 per share;
- 16,339,060 shares of Common Stock issuable upon the exercise of outstanding warrants to purchase Common Stock with a weighted average exercise price of \$0.55 per share; and
- 3,391,465 additional shares of the Common Stock reserved for future issuance under our equity incentive plans.

RISK FACTORS

This prospectus summary highlights information contained elsewhere in this prospectus. This summary does not contain all of the information that you should consider before deciding to invest in our Common Stock. You should read this entire prospectus and the documents incorporated by reference herein carefully, including the section entitled "Risk Factors" in this prospectus and under "Risk Factors" in our most recent Annual Report on Form 10-K, along with our consolidated financial statements and notes to those consolidated financial statements and the other information incorporated by reference in this prospectus, before deciding to invest in our Common Stock.

Risks Related to this Offering

The number of shares being registered for resale is significant in relation to the number of our outstanding shares of Common Stock.

We have filed a registration statement of which this prospectus forms a part to register the shares offered hereunder for sale into the public market by the Selling Stockholders. These shares represent a large number of shares of our outstanding shares of Common Stock, and if sold in the market all at once or at about the same time, could depress the market price of our Common Stock during the period the registration statement remains effective and could also affect our ability to raise equity capital.

Resales of our Common Stock in the public market by our stockholders as a result of this offering may cause the market price of our Common Stock to fall.

We are registering shares of Common Stock issuable upon the exercise of the New Warrants. Sales of substantial amounts of shares of our Common Stock in the public market, or the perception that such sales might occur, could adversely affect the market price of our Common Stock. The issuance of new shares of Common Stock could result in resales of our Common Stock by our current stockholders concerned about the potential ownership dilution of their holdings. Furthermore, in the future, we may issue additional shares of Common Stock or other equity or debt securities exercisable or convertible into Common Stock. Any such issuance could result in substantial dilution to our existing stockholders and could cause our stock price to decline.

Investors who buy shares at different times will likely pay different prices.

Investors who purchase shares in this offering at different times will likely pay different prices and so may experience different levels of dilution and different outcomes in their investment results.

Our management will have broad discretion over the use of the net proceeds from the cash exercise of the New Warrants by the Selling Stockholders, and you may not agree with how we use the proceeds, and the proceeds may not be invested successfully.

The Selling Stockholders will receive all of the proceeds of the sale of shares of Common Stock offered from time to time pursuant to this prospectus. Accordingly, we will not receive any proceeds from the sale of shares of Common Stock that may be sold from time to time pursuant to this prospectus; however, we will receive proceeds from the cash exercise of the New Warrants. We intend to use the proceeds from any cash exercise of the New Warrants for working capital and other general corporate purposes. However, we have not designated any portion of the net proceeds from the exercise of New Warrants to be used for any particular purpose. Accordingly, our management will have broad discretion as to the use of the net proceeds, if any, and you will be relying on the judgment of our management with regard to the use of these net proceeds, and you will not have the opportunity, as part of your investment decision, to assess whether the proceeds are being used appropriately. It is possible that, pending their use, we may invest the net proceeds in a way that does not yield a favorable, or any, return for our company. Our management's judgment may not result in positive returns on your investment, and you will not have the opportunity to evaluate the economic, financial or other information upon which our management bases its decisions.

We do not intend to pay dividends in the foreseeable future on our Common Stock.

We have never paid cash dividends on our Common Stock. We currently intend to retain our future earnings, if any, to finance the operation and growth of our business and currently do not plan to pay any cash dividends in the foreseeable future. If we do not pay dividends, our Common Stock may be less valuable because a return on your investment will only occur if the market price of our Common Stock price appreciates.

SPECIAL NOTE REGARDING FORWARD-LOOKING STATEMENTS

This prospectus and the documents we file with the SEC that are incorporated by reference herein and therein contain “forward-looking statements” within the meaning of Section 27A of the Securities Act, and Section 21E of the Exchange Act, these statements reflect our current views with respect to our ongoing and planned clinical trials, business strategy, business plan, financial performance and other future events. These statements include forward-looking statements both with respect to us, specifically, and the biotechnology sector, in general. We make these statements pursuant to the safe harbor provisions of the Private Securities Litigation Reform Act of 1995. Statements that include the words “expect,” “intend,” “plan,” “believe,” “project,” “estimate,” “may,” “should,” “anticipate,” “will” and similar statements of a future or forward-looking nature identify forward-looking statements for purposes of the federal securities laws or otherwise.

All forward-looking statements involve inherent risks and uncertainties, and there are or will be important factors that could cause actual results to differ materially from those indicated in these statements. We believe that these factors include, but are not limited to, those factors set forth under the caption “Risk Factors” in this prospectus and under the captions “Risk Factors,” “Business,” and “Management’s Discussion and Analysis of Financial Condition and Results of Operations,” in our most recent Annual Report on Form 10-K and our subsequent Quarterly Reports on Form 10-Q, all of which you should review carefully. Please consider our forward-looking statements in light of those risks as you read this prospectus. It is not possible for our management to predict all risks, nor can we assess the impact of all factors on our business or the extent to which any factor, or combination of factors, may cause actual results to differ materially from those contained in any forward-looking statements we may make. Given these uncertainties, you should not place undue reliance on these forward-looking statements.

You should not assume that the information contained in this prospectus is accurate as of any date other than the date on the front of this prospectus, or that any information incorporated by reference into this prospectus is accurate as of any date other than the date of the document so incorporated by reference. Except as required by law, we assume no obligation to update these forward-looking statements publicly, or to update the reasons actual results could differ materially from those anticipated in these forward-looking statements, even if new information becomes available in the future. Thus, you should not assume that our silence over time means that actual events are bearing out as expressed or implied in such forward-looking statements.

If one or more of these or other risks or uncertainties materializes, or if our underlying assumptions prove to be incorrect, actual results may vary materially from what we anticipate. All subsequent written and oral forward-looking statements attributable to us or individuals acting on our behalf are expressly qualified in their entirety by this Note. Before purchasing any shares of Common Stock, you should consider carefully all of the factors set forth or referred to in this prospectus that could cause actual results to differ.

USE OF PROCEEDS

The Selling Stockholders will receive all of the proceeds of the sale of shares of Common Stock offered from time to time pursuant to this prospectus. Accordingly, we will not receive any proceeds from the sale of shares of Common Stock that may be sold from time to time pursuant to this prospectus; however, we will receive proceeds from the cash exercise of the New Warrants. We intend to use the proceeds from any cash exercise of the New Warrants for working capital and other general corporate purposes.

We will bear the out-of-pocket costs, expenses and fees incurred in connection with the registration of shares of Common Stock to be sold by the Selling Stockholders pursuant to this prospectus. Other than registration expenses, the Selling Stockholders will bear any underwriting discounts, commissions, placement agent fees or other similar expenses payable with respect to sales of shares of Common Stock.

DIVIDEND POLICY

We have never declared or paid any cash dividends on shares of Common Stock and we do not currently intend to pay any cash dividends on our Common Stock in the foreseeable future. We expect to retain all available funds and future earnings, if any, to fund the development and growth of our business. Any future determination to pay dividends, if any, on our shares of Common Stock will be at the discretion of our board of directors and will depend on, among other factors, the terms of any outstanding preferred stock, our results of operations, financial condition, capital requirements and contractual restrictions.

DESCRIPTION OF CAPITAL STOCK

The following is a description of the material terms of our capital stock. This is a summary only and does not purport to be complete. It is subject to and qualified in its entirety by reference to our Articles of Incorporation and our Bylaws, each of which are incorporated by reference as an exhibit to the registration statement of which this prospectus forms a part. We encourage you to read our Certificate of Incorporation, our Bylaws and the applicable provisions of the Nevada Revised Statute (the "NRS"), for additional information.

Our authorized capital stock consists of: (i) 350,000,000 shares of Common Stock, par value \$0.001 per share; and (ii) 10,000,000 shares of preferred stock, par value \$0.001 per share.

Common Stock

Outstanding Shares. As of November 3, 2025, there were 33,739,643 shares of our Common Stock outstanding.

Voting Rights. The holders of the Common Stock are entitled to one vote for each share held of record on all matters submitted to a vote of the stockholders, including the election of directors, and do not have cumulative voting rights. Accordingly, the holders of a majority of the shares of the Common Stock entitled to vote in any election of directors can elect all of the directors standing for election.

Dividend Rights. Subject to preferences that may be applicable to any then outstanding preferred stock, the holders of Common Stock are entitled to receive dividends, if any, as may be declared from time to time by our board of directors out of legally available funds.

Liquidation Rights. In the event of our liquidation, dissolution or winding up, holders of the Common Stock will be entitled to share ratably in the net assets legally available for distribution to stockholders after the payment of all of our debts and other liabilities, subject to the satisfaction of any liquidation preference granted to the holders of any then outstanding shares of preferred stock.

Other Rights and Preferences. The holders of the Common Stock have no preemptive, conversion or subscription rights, and there are no redemption or sinking fund provisions applicable to the Common Stock. The rights, preferences and privileges of the holders of the Common Stock are subject to, and may be adversely affected by, the rights of the holders of shares of any series of our preferred stock that we may designate and issue in the future.

Fully Paid and Nonassessable. All of our outstanding shares of Common Stock are fully paid and nonassessable.

Anti-Takeover Effects of Nevada Law

The provisions of NRS, our Articles of Incorporation and our Bylaws described below may have the effect of delaying, deferring or discouraging another party from acquiring control of us.

Business Combinations

The "business combination" provisions of Sections 78.411 to 78.444, inclusive, of the NRS generally prohibit a Nevada corporation with at least 200 stockholders from engaging in various "combination" transactions with any interested stockholder for a period of two years after the date of the transaction in which the person became an interested stockholder, unless the transaction is approved by the board of directors prior to the date the interested stockholder obtained such status or the combination is approved by the board of directors and thereafter is approved at a meeting of the stockholders by the affirmative vote of stockholders representing at least 60% of the outstanding voting power held by disinterested stockholders, and extends beyond the expiration of the two-year period, unless:

- the combination was approved by the board of directors prior to the person becoming an interested stockholder or the transaction by which the person first became an interested stockholder was approved by the board of directors before the person became an interested stockholder or the combination is later approved by a majority of the voting power held by disinterested stockholders; or

- if the consideration to be paid by the interested stockholder is at least equal to the highest of: (a) the highest price per share paid by the interested stockholder within the two years immediately preceding the date of the announcement of the combination or in the transaction in which it became an interested stockholder, whichever is higher, (b) the market value per share of Common Stock on the date of announcement of the combination and the date the interested stockholder acquired the shares, whichever is higher, or (c) for holders of preferred stock, the highest liquidation value of the preferred stock, if it is higher.

A “combination” is generally defined to include mergers or consolidations or any sale, lease exchange, mortgage, pledge, transfer, or other disposition, in one transaction or a series of transactions, with an “interested stockholder” having: (a) an aggregate market value equal to 5% or more of the aggregate market value of the assets of the corporation, (b) an aggregate market value equal to 5% or more of the aggregate market value of all outstanding shares of the corporation, (c) 10% or more of the earning power or net income of the corporation, and (d) certain other transactions with an interested stockholder or an affiliate or associate of an interested stockholder.

In general, an “interested stockholder” is a person who, together with affiliates and associates, owns (or within two years, did own) 10% or more of a corporation’s voting stock. The statute could prohibit or delay mergers or other takeover or change in control attempts and, accordingly, may discourage attempts to acquire our company, even though such a transaction may offer our stockholders the opportunity to sell their stock at a price above the prevailing market price.

Control Share Acquisitions

The “control share” provisions of Sections 78.378 to 78.3793, inclusive, of the NRS apply to “issuing corporations” that are Nevada corporations with at least 200 stockholders, including at least 100 stockholders of record who are Nevada residents, and that conduct business directly or indirectly in Nevada. The control share statute prohibits an acquirer, under certain circumstances, from voting its shares of a target corporation’s stock after crossing certain ownership threshold percentages, unless the acquirer obtains approval of the target corporation’s disinterested stockholders. The statute specifies three thresholds: one-fifth or more but less than one-third, one-third but less than a majority, and a majority or more, of the outstanding voting power. Generally, once an acquirer crosses one of the above thresholds, those shares in an offer or acquisition and acquired within 90 days thereof become “control shares” and such control shares are deprived of the right to vote until disinterested stockholders restore the right. These provisions also provide that if control shares are accorded full voting rights and the acquiring person has acquired a majority or more of all voting power, all other stockholders who do not vote in favor of authorizing voting rights to the control shares are entitled to demand payment for the fair value of their shares in accordance with statutory procedures established for dissenters’ rights.

A corporation may elect to not be governed by, or “opt out” of, the control share provisions by making an election in its articles of incorporation or bylaws, provided that the opt-out election must be in place on the 10th day following the date an acquiring person has acquired a controlling interest, that is, crossing any of the three thresholds described above. We have not opted out of the control share statutes, and will be subject to these statutes if we are an “issuing corporation” as defined in such statutes.

The effect of the Nevada control share statutes is that the acquiring person, and those acting in association with the acquiring person, will obtain only such voting rights in the control shares as are conferred by a resolution of the stockholders at an annual or special meeting. The Nevada control share law, if applicable, could have the effect of discouraging takeovers of our company.

Articles of Incorporation and Bylaws

Our Articles of Incorporation and Bylaws provide that:

- the authorized number of directors is determined by our board of directors;

- directors may be removed only by the affirmative vote of the holders of at least a majority of our voting stock, whether for cause or without cause;
- our Bylaws may be amended or repealed by our board of directors or by the affirmative vote of our stockholders;
- special meetings of the stockholders may be called by Chairman of the board, if any, the Vice Chairman of the board, if any, or the President or upon request of stockholders owning a majority of the outstanding shares our board of directors may fill vacancies on the board of directors;

- our board of directors will be authorized to issue, without stockholder approval, preferred stock, the rights of which will be determined at the discretion of the board of directors and that, if issued, could operate as a “poison pill” to dilute the stock ownership of a potential hostile acquirer to prevent an acquisition that our board of directors does not approve;
- our stockholders do not have cumulative voting rights, and therefore our stockholders holding a majority of the shares of Common Stock outstanding will be able to elect all of our directors; and
- our stockholders must comply with advance notice provisions to bring business before or nominate directors for election at a stockholder meeting.

Potential Effects of Authorized but Unissued Stock

We have shares of Common Stock available for future issuance without stockholder approval. We may utilize these additional shares for a variety of corporate purposes, including future public offerings to raise additional capital, to facilitate corporate acquisitions or payment as a dividend on the capital stock.

The existence of unissued and unreserved Common Stock may enable our board of directors to issue shares to persons friendly to current management.

Limitations of Director Liability and Indemnification of Directors, Officers and Employees

NRS 78.138 provides that directors of a corporation is not individually liable to the corporation or its stockholders or creditors for any damages as a result of any act or failure to act in his or her capacity as a director or officer unless: (a) the presumption that directors and officers acted in good faith on an informed basis with a view toward the best interest of the corporation has been rebutted and (b) it is proven that:

- The director’s or officer’s act or failure to act constituted a breach of his or her fiduciary duties as a director or officer; and
- such breach involved intentional misconduct, fraud or a knowing violation of law.

Our Articles of Incorporation and Bylaws provide that we will indemnify our directors and officers to the fullest extent permitted by law and may indemnify employees and other agents. Our Articles of Incorporation also provide that we are obligated to advance expenses incurred by a director or officer in advance of the final disposition of any action or proceeding.

We have obtained a policy of directors’ and officers’ liability insurance.

We have entered into separate indemnification agreements with our directors and officers. These agreements, among other things, require us to indemnify our directors and officers for any and all expenses (including reasonable attorneys’ fees, retainers, court costs, transcript costs, fees of experts, witness fees, travel expenses, duplicating costs, printing and binding costs, telephone charges, postage, delivery service fees) judgments, fines and amounts paid in settlement actually and reasonably incurred by such directors or officers or on his or her behalf in connection with any action or proceeding arising out of their services as one of our directors or officers, or any of our subsidiaries or any other company or enterprise to which the person provides services at our request provided that such person follows the procedures for determining entitlement to indemnification and advancement of expenses set forth in the indemnification agreement. We believe that these bylaw provisions and indemnification agreements are necessary to attract and retain qualified persons as directors and officers.

The limitation of liability and indemnification provisions in our Articles of Incorporation and Bylaws may discourage stockholders from bringing a lawsuit against directors for breach of their fiduciary duties. They may also reduce the

likelihood of derivative litigation against directors and officers, even though an action, if successful, might provide a benefit to us and our stockholders. Our results of operations and financial condition may be harmed to the extent we pay the costs of settlement and damage awards against directors and officers pursuant to these indemnification provisions.

Insofar as indemnification for liabilities arising under the Securities Act may be permitted to directors, officers or persons controlling us, we have been informed that, in the opinion of the SEC, such indemnification is against public policy as expressed in the Securities Act and is therefore unenforceable.

At present, there is no pending litigation or proceeding involving any of our directors or officers as to which indemnification is required or permitted, and we are not aware of any threatened litigation or proceeding that may result in a claim for indemnification.

Listing of the Common Stock on the NYSE American LLC

Our Common Stock is listed for trading on the NYSE American LLC under the symbol "TOVX."

Transfer Agent and Registrar

The transfer agent and registrar for our Common Stock is Equiniti Trust Company.

Stock Options

As of November 3, 2025, we had options outstanding to purchase an aggregate of 1,116,101 shares of Common Stock that were issued under our equity compensation plans. As of November 3, 2025, there were 3,391,465 shares of Common Stock reserved for future issuance under our equity incentive plan.

DESCRIPTION OF THE WARRANT INDUCEMENT

On October 16, 2025, we entered into the Warrant Inducement Agreement with the Selling Stockholders, which were the beneficial owners of the Existing Warrants to purchase an aggregate of up to 8,092,280 shares of our Common Stock consisting of: (i) the September 2024 Warrants to purchase up to an aggregate of 1,345,000 shares of Common Stock, and (ii) the May 2025 Warrants to purchase up to an aggregate of 6,747,280 shares of our Common Stock. Pursuant to the Warrant Inducement Agreement, (1) the Selling Stockholders agreed to exercise for cash the Existing Warrants at a reduced exercise price of \$0.54 per share and (2) we agreed to issue to the Selling Stockholders the New Warrants to purchase up to an aggregate of 16,184,560 shares of Common Stock.

The issuance of the shares of Common Stock underlying the September 2024 Warrants were registered pursuant to our effective registration statement on Form S-1 (File No. 333-282024), which was initially filed with the SEC on September 10, 2024, as subsequently amended, and declared effective by the SEC on September 25, 2024 (the "September 2024 Registration Statement"), and the issuance of shares of Common Stock underlying the May 2025 Warrants were registered pursuant to an effective registration statement on Form S-1 (File No. 333-283722), which was initially filed with the SEC on December 10, 2024, as subsequently amended, and declared effective by the SEC on May 7, 2025 (the "May 2025 Registration Statement" and, together with the September 2024 Registration Statement, the "Registration Statements").

The transactions contemplated by the Warrant Inducement Agreement closed on October 17, 2025 (the "Closing Date"). We received aggregate gross proceeds of approximately \$4.4 million for the exercise of the Existing Warrants, before deducting placement agent fees and other expenses payable by us.

In consideration of the Selling Stockholders' agreement to exercise the Existing Warrants (the "Warrant Exercise"), at the reduced exercise price of \$0.54 per share, in accordance with the Warrant Inducement Agreement, we issued to the Selling Stockholder the New Warrants to purchase an aggregate of 16,184,560 shares of Common Stock, equal to 200% of the number of shares of Common Stock underlying the Existing Warrants, at an exercise price of \$0.54 per share. The shares of Common Stock issuable upon exercise of the New Warrants are sometimes hereinafter referred to collectively as the "New Warrant Shares." The New Warrants and the New Warrant Shares were initially issued in a private placement pursuant to the exemption provided in Section 4(a)(2) under the Securities Act and Rule 506(b) promulgated thereunder.

We agreed in the Warrant Inducement Agreement to file this registration statement with the SEC to register the resale of the New Warrant Shares (the "Resale Registration Statement") as soon as practicable (and in any event within 30 calendar days following the date of the Warrant Inducement Agreement), and to use commercially reasonable efforts to have this Resale Registration Statement declared effective by the SEC within 60 days following the date of the Warrant Inducement Agreement (or 90 days following the date of the Warrant Inducement Agreement in the event of a "limited review" or "full review" by the SEC), provided, however, that if the SEC is closed for operations due to a government shutdown, the applicable deadline for the Resale Registration Statement becoming effective shall be extended by the same amount of days that the SEC remains closed for operations, and to keep the Resale Registration Statement effective at all times until no holder of the New Warrants owns any New Warrants or New Warrant Shares.

Pursuant to the Warrant Inducement Agreement, we have agreed, with certain limited exceptions, for 45 days following the Closing Date, not to issue any shares of Common Stock or Common Stock Equivalents (as defined in the New Warrants) or to file any registration statement, subject to certain exceptions. We have also agreed not to enter into a Variable Rate Transaction (as defined in the Warrant Inducement Agreement) for a period of 90 days following the Closing Date, provided however that sales pursuant to an at-the-market are permitted to be made beginning five (5) days after Closing Date.

We expect to use the net proceeds from the Warrant Exercise for working capital.

A.G.P./Alliance Global Partners LP (“AGP”) served as our exclusive financial advisor in connection with the Warrant Exercise and other transactions described in the Warrant Inducement Agreement. Pursuant to the terms of an engagement letter, dated October 16, 2025, by and between us and AGP (the “Engagement Letter”), we agreed to pay to AGP a cash fee equal to 7.0% of the aggregate gross proceeds received from the Selling Stockholders upon exercise of the Existing Warrants and reimbursement of certain expenses.

Terms of the New Warrants

The New Warrants are exercisable on the date of Stockholder Approval (as defined below) and approval of the NYSE American in compliance with its rules and regulations and will expire on the fifth anniversary of the date of Stockholder Approval. If at any time after the date of Stockholder Approval a registration statement registering the issuance of the New Warrant Shares under the Securities Act is not effective or available, the holder may, in its sole discretion, elect to exercise the New Warrants through a cashless exercise, in which case the holder would receive upon such exercise the net number of shares of Common Stock determined according to the formula set forth in the New Warrants. “Stockholder Approval” means such approval from our stockholders of the full exercise of the New Warrants and the issuance of all of the New Warrant Shares upon the exercise thereof, it being understood that no portion of the New Warrants may be exercised until such approval from our stockholders is obtained. We agreed to use our reasonable best efforts to obtain all components of the Stockholder Approval. If we do not obtain all components of the Stockholder Approval at the first meeting of our stockholders called within sixty (60) days following the Closing Date, we will call a meeting every sixty (60) days thereafter to seek any remaining components of the Stockholder Approval until the earlier of the date on which Stockholder Approval is obtained or the New Warrants are no longer outstanding.

The exercise price of the New Warrants, and the number of New Warrant Shares, will be subject to adjustment in the event of any stock dividend or split, recapitalization, reorganization or similar transaction, as described in the New Warrants.

A holder of New Warrants will not have the right to exercise any portion of the New Warrants if the holder (together with its affiliates) would beneficially own in excess of 4.99% (or, upon election of the holder, 9.99%) of the number of shares of Common Stock outstanding immediately after giving effect to the exercise, as such percentage ownership is determined in accordance with the terms of the New Warrants. However, a holder may increase or decrease such percentage, provided that any increase will not be effective until the 61st day after such election.

In the event of a Fundamental Transaction (as such term is defined in the New Warrants), then the successor entity will succeed to, and be substituted for the us, and may exercise every right and power that we may exercise and will assume all of our obligations under the New Warrants with the same effect as if such successor entity had been named in the warrant itself. If holders of Common Stock are given a choice as to the securities, cash or property to be received in a Fundamental Transaction, then a holder of the New Warrants shall be given the same choice as to the consideration it receives upon any exercise of the New Warrants following such Fundamental Transaction. In addition, the successor entity, at the request of holders of New Warrants, will be obligated to purchase any unexercised portion of the New Warrants in accordance with the terms thereof. Notwithstanding the foregoing, in the event of a Fundamental Transaction, the holders of the New Warrants have the right to require us or a successor entity to purchase the New Warrant for cash in the amount of the Black Scholes Value (as defined in the New Warrant) of the unexercised portion of the New Warrants concurrently with or within 30 days following the consummation of a Fundamental Transaction. However, in the event of a Fundamental Transaction which is not in our control, including a Fundamental Transaction not approved by our board of directors, the holders of the New Warrants will only be entitled to receive from the us or our successor entity, as of the date of consummation of such Fundamental Transaction, the same type or form of consideration (and in the same proportion), at the Black Scholes Value of the unexercised portion of the New Warrant that is being offered and paid to the holders of Common Stock in connection with the Fundamental Transaction, whether that consideration is in the form of cash, stock or any combination of cash and stock, or whether the holders of Common Stock are given the choice to receive alternative forms of consideration in connection with the Fundamental Transaction.

Except as otherwise provided in the New Warrants or by virtue of such holder's ownership of shares of Common Stock, the holder of a New Warrants will not have the rights or privileges of a holder of Common Stock, including any voting rights, until the holder exercises such New Warrant.

The foregoing summaries of the Warrant Inducement, including the securities to be issued in connection therewith, the Warrant Inducement Agreement and the New Warrants do not purport to be complete and are qualified in their entirety by reference to the definitive transaction documents, copies of which are attached as Exhibits 10.1 and 4.1, respectively, to our Current Report on Form 8-K filed with the SEC on October 17, 2025, and which are each incorporated herein by reference.

SELLING STOCKHOLDERS

The shares of Common Stock being offered by the Selling Stockholders are those issuable to the Selling Stockholders upon exercise of the New Warrants. For additional information regarding the issuances of the New Warrants, see “Description of the Warrant Inducement” elsewhere in this prospectus. We are registering the shares of Common Stock in order to permit the Selling Stockholders to offer the shares of Common Stock for resale from time to time, after Stockholder Approval has been obtained. Except for the entry into the Warrant Inducement Agreement, ownership of the New Warrants and the other securities (if any) listed in the table below, the Selling Stockholders have not had any material relationship with us within the past three (3) years, except that they acquired securities from us in the following transactions: (i) on September 27, 2024, we consummated a public offering of an aggregate of (a) 918,600 shares of Common Stock, (b) pre-funded warrants to purchase up to 510,000 shares of Common Stock, and (c) common stock purchase warrants to purchase up to 1,428,600 shares of Common Stock, and (ii) on May 8, 2025, we consummated a public offering of an aggregate of (a) 1,990,900 shares of Common Stock, (b) pre-funded warrants to purchase up to 4,827,280 shares of Common Stock, and (c) common stock purchase warrants to purchase up to 6,818,180 shares of Common Stock.

The table below lists the Selling Stockholders and other information regarding the beneficial ownership of the shares of Common Stock by each of the Selling Stockholders. Beneficial ownership is determined in accordance with the rules of the SEC and includes voting or investment power with respect to shares of our Common Stock or if a person has the right to acquire beneficial ownership of our Common Stock within 60 days.

Unless otherwise indicated below, to our knowledge, the Selling Stockholders named in the table below have sole voting and investment power with respect to the shares of Common Stock beneficially owned by them. The number of shares of Common Stock disclosed as beneficially owned in the table below includes (i) all shares of our Common Stock held by such Selling Stockholder as of November 3, 2025, and (ii) all shares of our Common Stock issuable to such Selling Stockholder upon the exercise of New Warrants. The inclusion of any shares in this table does not constitute an admission of beneficial ownership by the person named below.

The second column in the table below titled “Number of Shares of Common Stock Beneficially Owned Prior to Offering” lists the number of shares of Common Stock beneficially owned by the Selling Stockholders, based on their respective ownership of shares of our Common Stock and the New Warrants, as of November 3, 2025, assuming exercise of the New Warrants held by the Selling Stockholders on that date, without regard to any limitations on exercises. As of November 3, 2025, we had 33,729,643 shares of our Common Stock outstanding.

The third column in the table below titled “Maximum Number of Shares of Common Stock to be Sold in this Offering” lists the maximum number of shares of Common Stock to be sold by this prospectus by the Selling Stockholders.

In accordance with the terms of the Warrant Inducement Agreement with the Selling Stockholders, this prospectus generally covers the resale of the maximum number of shares of Common Stock issuable upon exercise of the New Warrants, determined as if the outstanding New Warrants were exercised in full as of the trading day immediately preceding the date this registration statement was initially filed with the SEC, each as of the trading day immediately preceding the applicable date of determination and all subject to adjustment as provided in the registration right agreement, without regard to any limitations on the exercise of the New Warrants. The fourth and fifth columns in the table below titled “Number of Shares of Common Stock Beneficially Owned After Offering” and “Percentage of Shares Beneficially Owned After Offering”, assume the sale of all of the shares of Common Stock offered by the Selling Stockholders pursuant to this prospectus.

Under the terms of the New Warrants, the Selling Stockholders may not exercise the New Warrants to the extent such exercise would cause such Selling Stockholder, together with its affiliates and attribution parties, to beneficially own a number of shares of Common Stock that would exceed 4.99%, of our then outstanding Common Stock following such exercise, excluding for purposes of such determination shares of Common Stock issuable upon exercise of such New Warrants which have not been exercised. The number of shares in the second and third columns do not reflect this

limitation. The Selling Stockholders may sell all, some or none of their shares in this offering (assuming Stockholder Approval is obtained and NYSE American approval is obtained). See “Plan of Distribution” section of this prospectus for more information.

Name of Selling Stockholders	Number of Shares of Common Stock Beneficially Owned Prior to Offering⁽¹⁾	Maximum Number of Shares of Common Stock to be Sold in this Offering	Number of Shares of Common Stock Beneficially Owned After Offering	Percentage of Shares Beneficially Owned after Offering⁽¹⁾
Anson Investments Master Fund LP ⁽¹⁾	3,193,638	3,193,638	—	—
Anson East Master Fund LP ⁽²⁾	740,002	740,002	—	—
Armistice Capital, LLC ⁽³⁾	4,503,640	4,503,640	—	—
CVI Investments, Inc. ⁽⁴⁾	570,000	570,000	—	—
Funds managed by Empery Asset Management, LP ⁽⁵⁾	3,363,640	3,363,640	—	—
Hudson Bay Master Fund Ltd. ⁽⁶⁾	3,363,640	3,363,640	—	—
Intracoastal Capital LLC ⁽⁷⁾	300,000	300,000	—	—
Robert Forster ⁽⁸⁾	150,000	150,000	—	—

* less than 1%

- (1) Consists of 3,193,638 shares of our Common Stock issuable upon exercise of the New Warrants.

Anson Advisors Inc. and Anson Funds Management LP, the Co-Investment Advisers of Anson Investments Master Fund LP (“Anson Investments”), hold voting and dispositive power over our securities held by Anson Investments. Tony Moore is the managing member of Anson Management GP LLC, which is the general partner of Anson Funds Management LP. Moez Kassam and Amin Nathoo are directors of Anson Advisors Inc. Mr. Moore, Mr. Kassam and Mr. Nathoo each disclaim beneficial ownership of these securities except to the extent of their pecuniary interest therein. The principal business address of Anson Investments is Maples Corporate Services Limited, PO Box 309, Ugland House, Grand Cayman, KY1-1104, Cayman Islands.

- (2) Consists of 740,002 shares of our Common Stock issuable upon exercise of the New Warrants.

Anson Advisors Inc. and Anson Funds Management LP, the Co-Investment Advisers of Anson East Master Fund LP (“Anson East”), hold voting and dispositive power over our securities held by Anson East. Tony Moore is the managing member of Anson Management GP LLC, which is the general partner of Anson Funds Management LP. Moez Kassam and Amin Nathoo are directors of Anson Advisors Inc. Mr. Moore, Mr. Kassam and Mr. Nathoo each disclaim beneficial ownership of these securities except to the extent of their pecuniary interest therein. The principal business address of Anson East is Maples Corporate Services Limited, PO Box 309, Ugland House, Grand Cayman, KY1-1104, Cayman Islands.

- (3) Consists of 4,503,640 shares of our Common Stock issuable upon exercise of the New Warrants.

The securities are directly held by Armistice Capital Master Fund Ltd., a Cayman Islands exempted company (the “Master Fund”), and may be deemed to be beneficially owned by: (i) Armistice Capital, LLC (“Armistice Capital”), as the investment manager of the Master Fund; and (ii) Steven Boyd, as the Managing Member of Armistice Capital. The warrants are subject to a beneficial ownership limitation of 4.99%, which such limitation restricts the Selling Stockholder from exercising that portion of the warrants that would result in the Selling Stockholder and its affiliates owning, after exercise, a number of shares of Common Stock in excess of the beneficial ownership

- (5) Consists of: (i) 1,496,648 shares of our Common Stock issuable upon exercise of the New Warrants held by Empery Asset Master, LTD (“EAM”), (ii) 470,776 shares of our Common Stock issuable upon exercise of the New Warrants held by Empery Tax Efficient, LP (“ETE”), and (iii) 1,396,216 shares of our Common Stock issuable upon exercise of the New Warrants held by Empery Tax Efficient III, LP (“ETE III,” and collectively with EAM and ETE, the “Empery Funds”).

Empery Asset Management LP, the authorized agent of the Empery Funds, has discretionary authority to vote and dispose of the shares held by the Empery Funds and may be deemed to be the beneficial owner of these shares. Martin Hoe and Ryan Lane, in their capacity as investment managers of Empery Asset Management LP, may also be deemed to have investment discretion and voting power over the shares held by the Empery Funds. Each of the Empery Funds, Mr. Hoe and Mr. Lane each disclaim any beneficial ownership of these shares. The address of Empery Asset Management, LP is 1 Rockefeller Plaza, Suite 1205, New York, NY 10020.

- (6) Consists of 3,363,640 shares of our Common Stock issuable upon exercise of the New Warrants. Hudson Bay Capital Management LP, the investment manager of Hudson Bay Master Fund Ltd., has voting and investment power over these securities. Sander Gerber is the managing member of Hudson Bay Capital GP LLC, which is the general partner of Hudson Bay Capital Management LP. Each of Hudson Bay Master Fund Ltd. and Sander Gerber disclaims beneficial ownership over these securities. The mailing address is c/o Hudson Bay Capital Management LP, 290 Harbor Drive, Floor 3, Stamford, CT 06902.
- (7) Consists of 300,000 shares of our Common Stock issuable upon exercise of the New Warrants. Mitchell P. Kopin (“Mr. Kopin”) and Daniel B. Asher (“Mr. Asher”), each of whom are managers of Intracoastal Capital LLC (“Intracoastal”), have shared voting control and investment discretion over the securities reported herein that are held by Intracoastal. As a result, each of Mr. Kopin and Mr. Asher may be deemed to have beneficial ownership (as determined under Section 13(d) of the Exchange Act) of the securities reported herein that are held by Intracoastal. The business address of Intracoastal is 245 Palm Trail, Delray Beach, FL 33483
- (8) Consists of 150,000 shares of our Common Stock issuable upon exercise of the New Warrants. Robert Forster has sole voting control and investment power over the securities reported herein. The address of Mr. Forster is 54 Deepdale Drive, Great Neck, NY 11021.

PLAN OF DISTRIBUTION

Each Selling Stockholders of the securities and any of its pledgees, assignees and successors-in-interest may, from time to time, sell any or all of their securities covered hereby on the NYSE American or any other stock exchange, market or trading facility on which the securities are traded or in private transactions. These sales may be at fixed or negotiated prices. The Selling Stockholders may use any one or more of the following methods when selling securities:

- ordinary brokerage transactions and transactions in which the broker-dealer solicits purchasers;
- block trades in which the broker-dealer will attempt to sell the securities as agent but may position and resell a portion of the block as principal to facilitate the transaction;
- purchases by a broker-dealer as principal and resale by the broker-dealer for its account;
- an exchange distribution in accordance with the rules of the applicable exchange;
- privately negotiated transactions;
- settlement of short sales;
- in transactions through broker-dealers that agree with the Selling Stockholders to sell a specified number of such securities at a stipulated price per security;
- through the writing or settlement of options or other hedging transactions, whether through an options exchange or otherwise;
- a combination of any such methods of sale; or
- any other method permitted pursuant to applicable law.

The Stockholders may also sell securities under Rule 144 or any other exemption from registration under the Securities Act, if available, rather than under this prospectus.

Broker-dealers engaged by the Selling Stockholders may arrange for other brokers-dealers to participate in sales. Broker-dealers may receive commissions or discounts from the Selling Stockholders (or, if any broker-dealer acts as agent for the purchaser of securities, from the purchaser) in amounts to be negotiated, but, except as set forth in a supplement to this prospectus, in the case of an agency transaction not in excess of a customary brokerage commission in compliance with FINRA Rule 2121; and in the case of a principal transaction a markup or markdown in compliance with FINRA Rule 2121.

In connection with the sale of the securities or interests therein, the Selling Stockholders may enter into hedging transactions with broker-dealers or other financial institutions, which may in turn engage in short sales of the securities in the course of hedging the positions they assume. The Selling Stockholders may also sell securities short and deliver these securities to close out their short positions, or loan or pledge the securities to broker-dealers that in turn may sell these securities. The Selling Stockholders may also enter into option or other transactions with broker-dealers or other financial institutions or create one or more derivative securities which require the delivery to such broker-dealer or other financial institution of securities offered by this prospectus, which securities such broker-dealer or other financial institution may resell pursuant to this prospectus (as supplemented or amended to reflect such transaction).

The Selling Stockholders and any broker-dealers or agents that are involved in selling the securities may be deemed to be "underwriters" within the meaning of the Securities Act in connection with such sales. In such event, any commissions received by such broker-dealers or agents and any profit on the resale of the securities purchased by them

may be deemed to be underwriting commissions or discounts under the Securities Act. Each Selling Stockholders has informed us that it does not have any written or oral agreement or understanding, directly or indirectly, with any person to distribute the securities.

We are required to pay certain fees and expenses incurred by us incident to the registration of the securities.

We agreed to keep this prospectus effective until the date on which the New Warrants have been sold. Under applicable rules and regulations under the Exchange Act, any person engaged in the distribution of the resale securities may not simultaneously engage in market making activities with respect to the Common Stock for the applicable restricted period, as defined in Regulation M, prior to the commencement of the distribution. In addition, the Selling Stockholders will be subject to applicable provisions of the Exchange Act and the rules and regulations thereunder, including Regulation M, which may limit the timing of purchases and sales of the Common Stock by the Selling Stockholders or any other person. We will make copies of this prospectus available to the Selling Stockholders and have informed them of the need to deliver a copy of this prospectus to each purchaser at or prior to the time of the sale (including by compliance with Rule 172 under the Securities Act).

LEGAL MATTERS

Blank Rome LLP, New York, New York is representing us in connection with the offering. The validity of the shares of Common Stock offered hereby will be passed upon for us by Parsons Behle & Latimer, Reno, Nevada.

As of the date of this prospectus, an attorney of Blank Rome LLP beneficially owns securities exercisable to purchase shares of Common Stock that represent less than 1% of our outstanding shares of Common Stock.

EXPERTS

The consolidated financial statements of Theriva Biologics, Inc. (the Company) as of December 31, 2024 and 2023 and for each of the two years in the period ended December 31, 2024 incorporated by reference in this prospectus and in the registration statement have been so incorporated in reliance on the report of BDO USA, P.C, an independent registered public accounting firm, given on the authority of said firm as experts in auditing and accounting. The report on the consolidated financial statements contains an explanatory paragraph regarding the Company's ability to continue as a going concern.

WHERE YOU CAN FIND MORE INFORMATION

This prospectus is part of a registration statement we filed with the SEC. This prospectus does not contain all of the information set forth in the registration statement and the exhibits to the registration statement. For further information with respect to us and the securities we are offering under this prospectus, we refer you to the registration statement and the exhibits and schedules filed as a part of the registration statement. Neither we nor any agent, underwriter or dealer has authorized any person to provide you with different information. We are not making an offer of these securities in any state where the offer is not permitted. You should not assume that the information in this prospectus is accurate as of any date other than the date on the front page of this prospectus, regardless of the time of delivery of this prospectus or any sale of the securities offered by this prospectus.

We file annual, quarterly and current reports, proxy statements and other information with the SEC. Our SEC filings are available to the public at the SEC's website at www.sec.gov. Additional information about Synthetic Biologics, Inc. is contained at our website, www.syntheticbiologics.com. Information on our website is not incorporated by reference into this prospectus. We make available on our website our SEC filings as soon as reasonably practicable after those reports are filed with the SEC. The following Corporate Governance documents are also posted on our website: Code of Ethics and the Charters for the Audit Committee, Compensation Committee and Nominating and Governance Committee of the Board of Directors.

INCORPORATION OF CERTAIN DOCUMENTS BY REFERENCE

The SEC allows us to "incorporate by reference" information from other documents that we file with it, which means that we can disclose important information to you by referring you to those documents. The information incorporated by reference is considered to be part of this prospectus. Information in this prospectus supersedes information incorporated by reference that we filed with the SEC prior to the date of this prospectus.

We incorporate by reference into this prospectus, and the registration statement of which this prospectus is a part, the information or documents listed below that we have filed with the SEC (Commission File No. 001-12584):

- Our Annual Report on Form 10-K for the fiscal year ended December 31, 2024 filed with the SEC on March 6, 2025;
- Our Quarterly Report on Form 10-Q for the quarter ended March 31, 2025 filed with the SEC on May 14, 2025;

- Our Quarterly Report on Form 10-Q for the quarter ended June 30, 2025 filed with the SEC on August 11, 2025;
- Our Current Reports on Form 8-K filed with the SEC on March 31, 2025 (other than as set forth therein), May 7, 2025 (other than as set forth therein), May 8, 2025, June 2, 2025 (other than as set forth therein), June 20, 2025, June 25, 2025, September 4, 2025, October 2, 2025, October 6, 2025 (other than as set forth therein), October 8, 2025, October 14, 2025 (other than as set forth therein), October 17, 2025, October 20, 2025, October 24, 2025 and October 29, 2025; and
- The description of our Common Stock set forth in (i) our registration statement on Form 8-A12B, filed with the SEC on June 20, 2007 (File No. 001-12584) and (ii) Exhibit 4.2—Description of Securities of Theriva Biologics, Inc. to our Annual Report on Form 10-K for the fiscal year ended December 31, 2024.

We also incorporate by reference any future filings (other than current reports furnished under Item 2.02 or Item 7.01 of Form 8-K and exhibits filed on such form that are related to such items unless such Form 8-K expressly provides to the contrary) made with the SEC pursuant to Sections 13(a), 13(c), 14 or 15(d) of the Exchange Act, including those made (i) on or after the date of the initial filing of the registration statement of which this prospectus forms a part and prior to effectiveness of such registration statement, and (ii) on or after the date of this prospectus but prior to the termination of the offering (i.e., until the earlier of the date on which all of the securities registered hereunder have been sold or the registration statement of which this prospectus forms a part has been withdrawn). Information in such future filings updates and supplements the information provided in this prospectus. Any statements in any such future filings will automatically be deemed to modify and supersede any information in any document we previously filed with the SEC that is incorporated or deemed to be incorporated herein by reference to the extent that statements in the later filed document modify or replace such earlier statements.

We will furnish without charge to each person, including any beneficial owner, to whom a prospectus is delivered, upon written or oral request, a copy of any or all of the documents incorporated by reference into this prospectus but not delivered with the prospectus, including exhibits that are specifically incorporated by reference into such documents. You should direct any requests for documents to:

Theriva Biologics, Inc.
9605 Medical Center Drive, Suite 270
Rockville, Maryland 20850
Telephone: (301) 417-4364
Attention: Corporate Secretary

You may also access these documents, free of charge, on the SEC's website at www.sec.gov or on our website at <https://ir.therivabio.com/sec-filings>. The information contained in, or that can be accessed through, our website is not incorporated by reference in, and is not part of, this prospectus.

In accordance with Rule 412 of the Securities Act, any statement contained in a document incorporated by reference herein shall be deemed modified or superseded to the extent that a statement contained herein or in any other subsequently filed document which also is or is deemed to be incorporated by reference herein modifies or supersedes such statement.

You should rely only on information contained in, or incorporated by reference into, this prospectus. We have not authorized anyone to provide you with information different from that contained in this prospectus or incorporated by reference into this prospectus. We are not making offers to sell the securities in any jurisdiction in which such an offer or solicitation is not authorized or in which the person making such offer or solicitation is not qualified to do so or to anyone to whom it is unlawful to make such an offer or solicitation.

**16,184,560 Shares
of Common Stock**

PROSPECTUS

December 2, 2025

EXHIBIT ST-2

Nejmeh v. Theriva Biologics, Inc., Case No. 3:26-cv-00705 (D. Nev.)

Securities Purchase Agreement, Exhibit 10.1 to Theriva Biologics, Inc. Form 8-K filed May 8, 2025

SEC accession no. 0001104659-25-046170

https://www.sec.gov/Archives/edgar/data/894158/000110465925046170/tm2514447d1_ex10-1.htm

Cited passages: Section 3.1(x), “Application of Takeover Protections”; Section 4.5, “Stockholder Rights Plan.”

Yellow highlighting has been added by Plaintiff to mark the passages cited in Plaintiff’s Supplemental Summary. The text of the filing is otherwise reproduced as published on the SEC’s EDGAR system.

SECURITIES PURCHASE AGREEMENT

This SECURITIES PURCHASE AGREEMENT (this “Agreement”) is dated as of May 7, 2025, between Theriva Biologics, Inc., a Nevada corporation (the “Company”), and each purchaser identified on the signature pages hereto (each, including its successors and assigns, a “Purchaser” and collectively the “Purchasers”).

WHEREAS, subject to the terms and conditions set forth in this Agreement and pursuant to an effective registration statement under the Securities Act (as defined below), the Company desires to issue and sell to each Purchaser, and each Purchaser, severally and not jointly, desires to purchase from the Company, securities of the Company as more fully described in this Agreement.

NOW, THEREFORE, IN CONSIDERATION of the mutual covenants contained in this Agreement, and for other good and valuable consideration the receipt and adequacy of which are hereby acknowledged, the Company and each Purchaser agree as follows:

Section 1. DEFINITIONS

1.1 Definitions. In addition to the terms defined elsewhere in this Agreement, for all purposes of this Agreement, the following terms have the meanings set forth in this Section 1.1:

“Acquiring Person” shall have the meaning ascribed to such term in Section 4.5.

“Action” shall have the meaning ascribed to such term in Section 3.1(j).

“Affiliate” means any Person that, directly or indirectly through one or more intermediaries, controls or is controlled by or is under common control with a Person as such terms are used in and construed under Rule 405 under the Securities Act.

“Board of Directors” means the board of directors of the Company.

“Business Day” means any day other than Saturday, Sunday, or other day on which banking institutions in the State of New York are authorized or required by law to remain closed.

“Closing” means the closing of the purchase and sale of the Securities pursuant to Section 2.1.

“Closing Date” means the Trading Day on which all of the Transaction Documents have been executed and delivered by the applicable parties thereto, and all conditions precedent to (i) the Purchasers’ obligations to pay the Subscription Amount at the Closing and (ii) the Company’s obligations to deliver the Securities, in each case, at the Closing have been satisfied or waived, but in no event later than the second (2nd) Trading Day following the date hereof.

“Commission” means the United States Securities and Exchange Commission.

“Common Stock” means the common stock of the Company, par value \$0.001 per share.

“Common Stock Equivalents” means any securities of the Company or the Subsidiaries which would entitle the holder thereof to acquire at any time Common Stock, including, without limitation, any debt, preferred stock, right, option, warrant or other instrument that is at any time convertible into or exercisable or exchangeable for, or otherwise entitles the holder thereof to receive, Common Stock.

“Common Warrants” means the common warrants delivered to the Purchasers at Closing in accordance with Section 2.2(a) hereof, which common warrants shall be exercisable immediately upon issuance and may be exercised

during a period of five years commencing from their issuance, in the form of Exhibit A attached hereto.

"Company Counsel" means Blank Rome LLP, with offices located at 1271 Avenue of The Americas, New York, NY 10020.

"Company Nevada Counsel" means Parsons Behle & Latimer, with offices located at 50 West Liberty Street, Suite 750, Reno, Nevada 89501.

"Disclosure Schedules" means the Disclosure Schedules of the Company delivered concurrently herewith.

"Disclosure Time" means, (i) if this Agreement is signed on a day that is not a Trading Day or after 9:00 a.m. (New York City time) and before midnight (New York City time) on any Trading Day, 9:01 a.m. (New York City time) on the Trading Day immediately following the date hereof, unless otherwise instructed as to an earlier time by the Placement Agent, and (ii) if this Agreement is signed between midnight (New York City time) and 9:00 a.m. (New York City time) on any Trading Day, no later than 9:01 a.m. (New York City time) on the date hereof, unless otherwise instructed as to an earlier time by the Placement Agent.

"DVP" shall have the meaning ascribed to such term in Section 2.1(v).

"Evaluation Date" shall have the meaning ascribed to such term in Section 3.1(r).

"Exchange Act" means the Securities Exchange Act of 1934, as amended, and the rules and regulations promulgated thereunder.

"Exempt Issuance" means the issuance of (a) shares of Common Stock, restricted stock units or options, including the shares of Common Stock underlying the restricted stock units or options, to employees, officers, directors or consultants of the Company or its subsidiary pursuant to any stock or option plan or arrangement duly adopted for such purpose by a majority of the non-employee members of the Board of Directors or a majority of the members of a committee of non-employee directors established for such purpose for services rendered to the Company provided that such securities issued to consultants are issued as "restricted securities" (as defined in Rule 144) and carry no registration rights that require or permit the filing of any registration statement in connection therewith during the prohibition period in Section 4.10(a) herein, (b) securities upon the exercise or exchange of or conversion of any Securities issued hereunder and/or other securities exercisable or exchangeable for or convertible into shares of Common Stock issued and outstanding on the date of this Agreement, *provided* that such securities have not been amended since the date of this Agreement to increase the number of such securities or to decrease the exercise price, exchange price or conversion price of such securities (other than in connection with stock splits or combinations or anti-dilution provisions contained therein as disclosed in the SEC Reports and the Prospectus) or to extend the term of such securities, (c) securities pursuant to merger, acquisition or strategic transactions approved by a majority of the disinterested directors of the Company, *provided* that such securities are issued as "restricted securities" (as defined in Rule 144) and carry no registration rights that require or permit the filing of any registration statement in connection therewith during the prohibition period in Section 4.10(a) herein, and *provided further* that any such issuance shall only be to a Person that is, itself or through its subsidiaries, an operating company in a business synergistic with the business of the Company and in which the Company receives benefits in addition to any investment of funds, but shall not include a transaction in which the Company is issuing securities primarily for the purpose of raising capital or to an entity whose primary business is investing in securities, and (d) securities for settlement of outstanding payables or liabilities *provided* that such securities are issued as "restricted securities" (as defined in Rule 144) and carry no registration rights that require or permit the filing of any registration statement in connection therewith during the prohibition period in Section 4.10(a) herein.

"FCPA" means the Foreign Corrupt Practices Act of 1977, as amended.

"GAAP" means generally accepted accounting principles in the United States.

“General Disclosure Package” means the Preliminary Prospectus, the Prospectus, and the Registration Statement.

“Indebtedness” shall have the meaning ascribed to such term in Section 3.1(z).

"Intellectual Property Rights" shall have the meaning ascribed to such term in Section 3.1(p).

"Liens" means a lien, charge, pledge, security interest, encumbrance, right of first refusal, preemptive right or other restriction.

"Lock-Up Agreements" means each Lock-Up Agreement, dated as of the date hereof, by and between the Company and each of the directors, officers and 5% stockholders of the Company, in the form of Exhibit C attached hereto.

"Material Adverse Effect" shall have the meaning assigned to such term in Section 3.1(b).

"Material Permits" shall have the meaning ascribed to such term in Section 3.1(m).

"Per Share Purchase Price" equals \$1.10, subject to adjustment for reverse and forward share splits, share dividends, share combinations and other similar transactions of shares of Common Stock that occur between the date hereof and the Closing Date.

"Per Pre-Funded Warrant Purchase Price" equals \$0.001, subject to adjustment for reverse and forward share splits, share dividends, share combinations and other similar transactions relating to shares of Common Stock that occur after the date of this Agreement.

"Person" means an individual or corporation, partnership, trust, incorporated or unincorporated association, joint venture, limited liability company, joint stock company, government (or an agency or subdivision thereof) or other entity of any kind.

"Placement Agent" means A.G.P./Alliance Global Partners.

"Placement Agent Counsel" means Thompson Hine LLP with offices located at 300 Madison Ave, 27th Floor, New York, New York 10017.

"Pre-Funded Warrants" means, collectively, the warrants delivered to the Purchasers at Closing in accordance with Section 2.2(a) hereof, which Pre-Funded Warrants shall be exercisable immediately upon issuance and shall expire when exercised in full, in the form of Exhibit B attached hereto.

"Pre-Notice" shall have the meaning ascribed to such term in Section 4.16(b).

"Preliminary Prospectus" means the preliminary prospectus included in the Registration Statement at the time the Registration Statement is declared effective.

"Proceeding" means an action, claim, suit, investigation or proceeding (including, without limitation, an informal investigation or partial proceeding, such as a deposition), whether commenced or threatened.

"Prospectus" means the final prospectus filed pursuant to the Registration Statement.

"Purchaser Party" shall have the meaning ascribed to such term in Section 4.8.

"Registration Statement" means the effective registration statement with the Commission on Form S-1 (File No. 333-283722), as amended, which registers the sale of the Securities and includes any Rule 462(b) Registration Statement.

"Required Approvals" shall have the meaning ascribed to such term in Section 3.1(e).

“Rule 144” means Rule 144 promulgated by the Commission pursuant to the Securities Act, as such Rule may be amended or interpreted from time to time, or any similar rule or regulation hereafter adopted by the Commission having substantially the same purpose and effect as such Rule.

“Rule 424” means Rule 424 promulgated by the Commission pursuant to the Securities Act, as such Rule may be amended or interpreted from time to time, or any similar rule or regulation hereafter adopted by the Commission having substantially the same purpose and effect as such Rule.

“Rule 462(b) Registration Statement” means any registration statement prepared by the Company registering additional Securities, which was filed with the Commission on or prior to the date hereof and became automatically effective pursuant to Rule 462(b) promulgated by the Commission pursuant to the Securities Act.

“SEC Reports” shall have the meaning ascribed to such term in Section 3.1(h).

“Securities” means the Shares, the Warrants, and the Warrant Shares.

“Securities Act” means the Securities Act of 1933, as amended, and the rules and regulations promulgated thereunder.

“Shares” means the shares of Common Stock issued and issuable to each Purchaser pursuant to this Agreement.

“Short Sales” means all “short sales” as defined in Rule 200 of Regulation SHO under the Exchange Act (but shall not be deemed to include locating and/or borrowing shares of Common Stock).

“Subscription Amount” means, as to each Purchaser, the aggregate amount to be paid for shares of Common Stock, and Pre-Funded Warrants, purchased hereunder as specified below such Purchaser’s name on the signature page of this Agreement and next to the heading “Subscription Amount,” in United States dollars and in immediately available funds.

“Subsidiary” means any significant subsidiary of the Company as set forth in the Exhibit 21.1 to the Company’s Annual Report on Form 10-K for the year ended December 31, 2023 and shall, where applicable, also include any direct or indirect subsidiary of the Company formed or acquired after the date hereof.

“Trading Day” means a day on which the principal Trading Market is open for trading.

“Trading Market” means any of the following markets or exchanges on which the shares of Common Stock are listed or quoted for trading on the date in question: the NYSE American, The Nasdaq Capital Market, The Nasdaq Global Market, The Nasdaq Global Select Market, or the New York Stock Exchange (or any successors to any of the foregoing).

“Transaction Documents” means this Agreement, the Warrants, the Lock-Up Agreements and all exhibits and schedules thereto and hereto and any other documents or agreements executed in connection with the transactions contemplated hereunder.

“Variable Rate Transaction” shall have the meaning ascribed to such term in Section 4.10(b).

“Warrants” means, collectively, the Common Warrants and the Pre-Funded Warrants.

“Warrant Shares” means, collectively, the shares of Common Stock issuable upon exercise of the Warrants.

Section 2. PURCHASE AND SALE

2.1 Closing. On the Closing Date, upon the terms and subject to the conditions set forth herein, the Company agrees to sell, and the Purchasers, severally and not jointly, agree to purchase, (i) the number of shares of Common Stock set forth under the heading “Subscription Amount” on the Purchaser’s signature page hereto, at the Per Share Purchase Price, and

(ii) Common Warrants exercisable for shares of Common Stock as calculated pursuant to Section 2.2(a); *provided, however,* that, to the extent that a Purchaser determines, in its sole discretion, that such Purchaser (together with such Purchaser's Affiliates, and any Person acting as a group together with such Purchaser or any of such Purchaser's Affiliates) would beneficially own in excess of the Beneficial Ownership Limitation, or as such Purchaser may otherwise choose, in lieu of purchasing shares of Common Stock, such Purchaser may elect to purchase Pre-Funded Warrants in such manner to result in the full Subscription Amount being paid by such Purchaser to the Company. The "Beneficial Ownership Limitation" shall be 4.99% (or, at the election of the Purchaser, 9.99%) of the number of shares of Common Stock, in each case, outstanding immediately after giving effect to the issuance of the Securities on the Closing Date.

Each Purchaser's Subscription Amount as set forth on the signature page hereto executed by such Purchaser shall be made available for Delivery Versus Payment ("DVP") settlement with the Company or its designees. The Company shall deliver to each Purchaser its respective Shares and Warrants as determined pursuant to Section 2.2(a), and the Company and each Purchaser shall deliver the other items set forth in Section 2.2 at the Closing. Upon satisfaction of the covenants and conditions set forth in Sections 2.2 and 2.3, the Closing shall occur remotely via the exchange of documents and signatures or such other location as the parties shall mutually agree. Unless otherwise directed by the Placement Agent, settlement of the Shares shall occur via DVP (i.e., on the Closing Date, the Company shall issue the Shares registered in the Purchasers' names and addresses and released by the Depositary directly to the account(s) at the Placement Agent identified by each Purchaser; upon receipt of such Shares, the Placement Agent shall promptly electronically deliver such Shares to the applicable Purchaser, and payment therefor shall be made by the Placement Agent (or its clearing firm) by wire transfer to the Company). Notwithstanding anything herein to the contrary, if at any time on or after the time of execution of this Agreement by the Company and an applicable Purchaser through, and including the time immediately prior to the Closing (the "Pre-Settlement Period"), such Purchaser sells to any Person all, or any portion, of any Securities to be issued hereunder to such Purchaser at the Closing (collectively, the "Pre-Settlement Shares"), such Person shall, automatically hereunder (without any additional required actions by such Purchaser or the Company), be deemed to be a Purchaser under this Agreement unconditionally bound to purchase, and the Company shall be deemed unconditionally bound to sell, such Pre-Settlement Shares to such Person at the Closing; provided, that the Company shall not be required to deliver any Pre-Settlement Shares to such Purchaser prior to the Company's receipt of the Subscription Amount for such Pre-Settlement Shares hereunder; and provided, further, that the Company hereby acknowledges and agrees that the forgoing shall not constitute a representation or covenant by such Purchaser as to whether or not such Purchaser will elect to sell any Pre-Settlement Shares during the Pre-Settlement Period. The decision to sell any shares of Common Stock by such Purchaser shall solely be made at the time such Purchaser elects to effect any such sale, if any. Notwithstanding the foregoing, with respect to any Notice(s) of Exercise (as defined in the Prefunded Warrants) delivered on or prior to 9:00 a.m. (New York City time) on the Closing Date, which may be delivered at any time after the time of execution of the this Agreement, the Company agrees to deliver the Warrant Shares subject to such notice(s) by 4:00 p.m. (New York City time) on the Closing Date and the Closing Date shall be the Warrant Share Delivery Date (as defined in the Prefunded Warrants) for purposes hereunder. Notwithstanding anything to the contrary herein and a Purchaser's Subscription Amount set forth on the signature pages attached hereto, the number of Shares purchased by a Purchaser (and its Affiliates) hereunder shall not, when aggregated with all other shares of Common Stock owned by such Purchaser (and its Affiliates) at such time, result in such Purchaser beneficially owning (as determined in accordance with Section 13(d) of the Exchange Act) in excess of 9.99% of the then issued and outstanding Common Stock outstanding at the Closing (the "Beneficial Ownership Maximum"), and such Purchaser's Subscription Amount, to the extent it would otherwise exceed the Beneficial Ownership Maximum immediately prior to the Closing, shall be conditioned upon the issuance of Shares at the Closing to the other Purchasers signatory hereto. To the extent that a Purchaser's beneficial ownership of the Shares would otherwise be deemed to exceed the Beneficial Ownership Maximum, such Purchaser's Subscription Amount shall automatically be reduced as necessary in order to comply with this paragraph.

2.2 Deliveries.

- (a) On or prior to the Closing Date, the Company shall deliver or cause to be delivered to each Purchaser the following:
 - (i) this Agreement duly executed by the Company;
 - (ii) the Company's wire instructions, on Company letterhead and executed by the Company's Chief Executive Officer or Chief Financial Officer;
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- (iii) subject to the provision of Section 2.1 that settlement of the Shares shall occur via DVP, a copy of the irrevocable instructions to the Transfer Agent instructing the Transfer Agent to deliver on an expedited basis via The Depository Trust Company Deposit or Withdrawal at Custodian system shares of Common Stock equal to the portion of such Purchaser's Subscription Amount divided by the Per Share Purchase Price, registered in the name of such Purchaser;
 - (iv) for each Purchaser of Pre-Funded Warrants pursuant to Section 2.1, a Pre-Funded Warrant registered in the name of such Purchaser to purchase up to a number of shares of Common Stock equal to the portion of such Purchaser's Subscription Amount applicable to Pre-Funded Warrants divided by the sum of the Per Pre-Funded Warrant Purchase Price, subject to adjustment therein;
 - (v) the Preliminary Prospectus and the Prospectus (which may be delivered in accordance with Rule 172 under the Securities Act);
 - (vi) a Common Warrant with an exercise price equal to \$1.10, registered in the name of such Purchaser, to purchase up to a number of shares of Common Stock equal to 100% of such Purchaser's shares of Common Stock, or a Pre-Funded Warrants, as applicable, with an exercise price equal to \$0.001 per share, subject to adjustment therein;
 - (vii) the duly executed Lock-Up Agreements;
 - (viii) a legal opinion of Company Counsel, in form reasonably acceptable to the Placement Agent and the Purchasers; and
 - (ix) a legal opinion of Company Nevada Counsel, in form reasonably acceptable to the Placement Agent and the Purchasers.
- (b) On or prior to the Closing Date, each Purchaser shall deliver or cause to be delivered to the Company, the following:
- (i) this Agreement duly executed by such Purchaser; and
 - (ii) such Purchaser's Subscription Amount with respect to the Securities purchased by such Purchaser, which shall be made available for DVP settlement with the Company or its designees.

2.3 Closing Conditions.

- (a) The obligations of the Company hereunder in connection with the Closing are subject to the following conditions being met:
- (i) the accuracy in all material respects (or, to the extent representations or warranties are qualified by materiality or Material Adverse Effect, in all respects) when made and on the Closing Date of the representations and warranties of the Purchasers contained herein (unless as of a specific date therein in which case they shall be accurate as of such date in all material respects or to the extent qualified by materiality or Material Adverse Effect);
 - (ii) all obligations, covenants and agreements of each Purchaser required to be performed at or prior to the Closing Date shall have been performed; and

(iii) the delivery by each Purchaser of the items set forth in Section 2.2(b) of this Agreement.

- (b) The respective obligations of the Purchasers hereunder in connection with the Closing are subject to the following conditions being met:
- (i) the accuracy in all material respects (or, to the extent representations or warranties are qualified by materiality or Material Adverse Effect, in all respects) when made and on the Closing Date of the representations and warranties of the Company contained herein (unless as of a specific date therein in which case they shall be accurate as of such date);
 - (ii) all obligations, covenants and agreements of the Company required to be performed at or prior to the Closing Date shall have been performed;
 - (iii) the delivery by the Company of the items set forth in Section 2.2(a) of this Agreement;
 - (iv) there shall have been no Material Adverse Effect with respect to the Company since the date hereof; and
 - (v) from the date hereof to the Closing Date, trading in the Common Stock shall not have been suspended by the Commission or any Trading Market, and, at any time prior to the Closing Date, trading in securities generally as reported by Bloomberg L.P. shall not have been suspended or limited, nor shall a banking moratorium have been declared either by the United States or New York State authorities nor shall there have occurred after the date of this Agreement any material outbreak or escalation of hostilities or other national or international calamity of such magnitude in its effect on, or any material adverse change in, any financial market which, in each case, in the reasonable judgment of such Purchaser, makes it impracticable or inadvisable to purchase the Securities at the Closing.

Section 3.

REPRESENTATIONS AND WARRANTIES

3.1 Representations and Warranties of the Company. Except as set forth in the Disclosure Schedules, which Disclosure Schedules shall be deemed a part hereof and shall qualify any representation or otherwise made herein to the extent of the disclosure contained in the corresponding section of the Disclosure Schedules, the Company hereby makes the following representations and warranties to each Purchaser:

- (a) Subsidiaries. Except as disclosed in the SEC Reports, the Company owns, directly or indirectly, all of the capital shares or other equity interests of each Subsidiary, free and clear of any Liens, and all of the issued and outstanding shares of capital stock of each Subsidiary are validly issued and are fully paid, non-assessable and free of preemptive and similar rights to subscribe for or purchase securities.
- (b) Organization and Qualification. The Company and each of the Subsidiaries is an entity duly incorporated or otherwise organized, validly existing, and, if applicable under the laws of the jurisdiction in which they are formed, in good standing under the laws of the jurisdiction of its incorporation or organization, with the requisite power and authority to own and use its properties and assets and to carry on its business as currently conducted. Neither the Company nor any Subsidiary is in violation nor default of any of the provisions of its respective memorandum of association, articles of association, certificate or articles of incorporation, bylaws, operating agreement, or other organizational or charter documents that would constitute a Material Adverse Effect (defined below). Each of the Company and the Subsidiaries is duly qualified to conduct business and is in good standing (where applicable) as a foreign corporation or other entity in each jurisdiction in which the nature of the business conducted or property owned by it makes such qualification necessary, except where the failure to be so qualified or in good standing, as the case may be, could not have or reasonably be expected to result in: (i) a material adverse effect on the legality, validity or enforceability of any Transaction Document, (ii) material adverse effect on the results of

operations, assets, business, prospects or condition (financial or otherwise) of the Company and the Subsidiaries, taken as a whole, or (iii) a material adverse effect on the Company's ability to perform in any material respect on a timely basis its obligations under any Transaction Document (any of (i), (ii) or (iii), a "Material Adverse Effect") and no Proceeding has been instituted in any such jurisdiction revoking, limiting or curtailing or seeking to revoke, limit or curtail such power and authority or qualification.

- (c) Authorization; Enforcement. The Company has the requisite corporate power and authority to enter into and to consummate the transactions contemplated by this Agreement and each of the other Transaction Documents and otherwise to carry out its obligations hereunder and thereunder. The execution and delivery of this Agreement and each of the other Transaction Documents by the Company and the consummation by it of the transactions contemplated hereby and thereby have been duly authorized by all necessary action on the part of the Company and no further action is required by the Company, the Board of Directors, a committee of the Board of Directors or the Company's stockholders in connection herewith or therewith other than in connection with the Required Approvals. This Agreement and each other Transaction Document to which it is a party has been (or upon delivery will have been) duly executed by the Company and, when delivered in accordance with the terms hereof and thereof, will constitute the valid and binding obligation of the Company enforceable against the Company in accordance with its terms, except (i) as limited by general equitable principles and applicable bankruptcy, insolvency, reorganization, moratorium and other laws of general application affecting enforcement of creditors' rights generally, (ii) as limited by laws relating to the availability of specific performance, injunctive relief or other equitable remedies and (iii) insofar as indemnification and contribution provisions may be limited by applicable law.
- (d) No Conflicts. The execution, delivery and performance by the Company of this Agreement and the other Transaction Documents to which it is a party, the issuance and sale of the Securities and the consummation by it of the transactions contemplated hereby and thereby do not and will not (i) conflict with or violate any provision of the Company's or any Subsidiary's memorandum of association, articles of association, certificate or articles of incorporation, bylaws, operating agreement, or other organizational or charter documents, or (ii) conflict with, or constitute a default (or an event that with notice or lapse of time or both would become a default) under, result in the creation of any Lien upon any of the properties or assets of the Company or any Subsidiary, or give to others any rights of termination, amendment, anti-dilution or similar adjustments acceleration or cancellation (with or without notice, lapse of time or both) of, any agreement, credit facility, debt or other instrument (evidencing a Company or Subsidiary debt or otherwise) or other understanding to which the Company or any Subsidiary is a party or by which any property or asset of the Company or any Subsidiary is bound or affected, or (iii) subject to the Required Approvals, conflict with or result in a violation of any law, rule, regulation, order, judgment, injunction, decree or other restriction of any court or governmental authority to which the Company or a Subsidiary is subject (including federal and state securities laws and regulations), or by which any property or asset of the Company or a Subsidiary is bound or affected; except in the case of each of clauses (ii) and (iii), such as could not have or reasonably be expected to result in a Material Adverse Effect.
- (e) Filings, Consents and Approvals. The Company is not required to obtain any consent, waiver, authorization or order of, give any notice to, or make any filing or registration with, any court or other federal, state, local or other governmental authority or other Person in connection with the execution, delivery and performance by the Company of the Transaction Documents, other than: (i) the filings required pursuant to Section 4.4 of this Agreement, (ii) the filing with the Commission of the Prospectus, (iii) notices and/or application(s) to and approvals by each applicable Trading Market for the listing of the applicable Securities for trading thereon in the time and manner required thereby, and (iv) filings required by the Financial Industry Regulatory Authority ("FINRA") (collectively, the "Required Approvals").
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- (f) Issuance of the Securities; Registration. The Shares and Warrant Shares are duly authorized and, when issued and paid for in accordance with the applicable Transaction Documents, will be duly and validly issued, fully paid and non-assessable, free and clear of all Liens imposed by the Company. The Warrants are duly authorized and binding obligations of the Company under the law of the jurisdiction governing the Warrants, and, when issued in accordance with this Agreement, will be duly and validly issued, and free and clear of all Liens imposed by the Company. The Company has reserved from its duly authorized share capital the maximum number of shares of Common Stock issuable pursuant to this Agreement and the Warrants. The Company has prepared and filed the Registration Statement in conformity with the requirements of the Securities Act, which became effective on May [7], 2025, including the Prospectus, and such amendments and supplements thereto as may have been required to the date of this Agreement. The Registration Statement is effective under the Securities Act and no stop order preventing or suspending the effectiveness of the Registration Statement or suspending or preventing the use of the Preliminary Prospectus or the Prospectus has been issued by the Commission and no proceedings for that purpose have been instituted or, to the knowledge of the Company, are threatened by the Commission. The Company, if required by the rules and regulations of the Commission, shall file the Preliminary Prospectus or the Prospectus with the Commission pursuant to Rule 424(b). At the time the Registration Statement and any amendments thereto became effective as determined under the Securities Act, at the date of this Agreement and at the Closing Date, the Registration Statement and any amendments thereto conformed and will conform in all material respects to the requirements of the Securities Act and did not and will not contain any untrue statement of a material fact or omit to state any material fact required to be stated therein or necessary to make the statements therein not misleading; and the Prospectus and any amendments or supplements thereto, at the time the Preliminary Prospectus, the Prospectus or any amendment or supplement thereto was issued and at the Closing Date, conformed and will conform in all material respects to the requirements of the Securities Act and did not and will not contain an untrue statement of a material fact or omit to state a material fact necessary in order to make the statements therein, in the light of the circumstances under which they were made, not misleading.
- (g) Capitalization. The capitalization of the Company as of the date hereof is as set forth in the General Disclosure Package, and includes the number of shares of Common Stock owned beneficially, and of record, by Affiliates of the Company as of the date hereof. The Company has not issued any capital stock since its most recently filed report under the Exchange Act, other than pursuant to the exercise of employee stock options under the Company's stock option plans, the issuance of shares of Common Stock to employees pursuant to the Company's employee stock purchase plans and pursuant to the conversion and/or exercise of Common Stock Equivalents outstanding as of the date of the most recently filed report under the Exchange Act. No Person has any right of first refusal, preemptive right, right of participation, or any similar right to participate in the transactions contemplated by the Transaction Documents. As a result of the purchase and sale of the Securities, there are no outstanding options, warrants, scrip rights to subscribe to, calls or commitments of any character whatsoever relating to, or securities, rights or obligations convertible into or exercisable or exchangeable for, or giving any Person any right to subscribe for or acquire, any shares of Common Stock, or contracts, commitments, understandings or arrangements by which the Company or any Subsidiary is or may become bound to issue additional shares of Common Stock or Common Stock Equivalents. The issuance and sale of the Securities will not result in a right of any holder of Company securities to adjust the exercise, conversion, exchange or reset price under any of such securities. There are no outstanding securities or instruments of the Company or any Subsidiary with any provision that adjusts the exercise, conversion, exchange or reset price of such security or instrument upon an issuance of securities by the Company or any Subsidiary. There are no outstanding securities or instruments of the Company or any Subsidiary that contain any redemption or similar provisions, and there are no contracts, commitments, understandings or arrangements by which the Company or any Subsidiary is or may become bound to redeem a security of the Company or such Subsidiary. The Company does not have any share appreciation rights or "phantom share" plans or agreements or any similar plan or agreement. All of the outstanding shares of the Company are duly authorized, validly issued, fully paid and non-assessable, have been issued in compliance with all federal and state securities laws where applicable,

and none of such outstanding shares was issued in violation of any preemptive rights or similar rights to subscribe for or purchase securities. Except for the Required Approvals, no further approval or authorization of any stockholder, the Board of Directors, or others is required for the issuance and sale of the Securities. There are no stockholders agreements, voting agreements or other similar agreements with respect to the Company's share capital to which the Company is a party or, to the knowledge of the Company, between or among any of the Company's stockholders.

- (h) SEC Reports; Financial Statements. The Company has filed all reports, schedules, forms, statements and other documents required to be filed by the Company under the Securities Act and the Exchange Act, including pursuant to Section 13(a) or 15(d) thereof, for the one year preceding the date hereof (or such shorter period as the Company was required by law or regulation to file such materials) (the foregoing materials, including the exhibits thereto and documents incorporated by reference therein, together with the Preliminary Prospectus and the Prospectus, being collectively referred to herein as the “SEC Reports”) As of their respective dates, the SEC Reports complied in all material respects with the requirements of the Securities Act and the Exchange Act, as applicable, and none of the SEC Reports, when filed, contained any untrue statement of a material fact or omitted to state a material fact required to be stated therein or necessary in order to make the statements therein, in the light of the circumstances under which they were made, not misleading. The financial statements of the Company included in the SEC Reports comply in all material respects with applicable accounting requirements and the rules and regulations of the Commission with respect thereto as in effect at the time of filing. Such financial statements have been prepared in accordance with GAAP, except as may be otherwise specified in such financial statements or the notes thereto and except that unaudited financial statements may not contain all footnotes required by GAAP, and fairly present in all material respects the financial position of the Company and its consolidated Subsidiaries as of and for the dates thereof and the results of operations and cash flows for the periods then ended, subject, in the case of unaudited statements, to normal, immaterial, year-end audit adjustments.
- (i) Material Changes; Undisclosed Events, Liabilities or Developments. Since the date of the latest financial statements filed by the Company with the SEC and other than the reverse stock split effected August 26, 2024, (i) there has been no event, occurrence or development that has had or that could reasonably be expected to result in a Material Adverse Effect, (ii) the Company has not incurred any liabilities (contingent or otherwise) other than (A) trade payables and accrued expenses incurred in the ordinary course of business consistent with past practice and strategic acquisitions and (B) liabilities not required to be reflected in the Company’s financial statements pursuant to GAAP or disclosed in filings made with the Commission, (iii) the Company has not altered its method of accounting, (iv) the Company has not declared or made any dividend or distribution of cash or other property to its stockholders or purchased, redeemed or made any agreements to purchase or redeem any of its shares and (v) the Company has not issued any equity securities to any officer, director or Affiliate, except pursuant to existing Company equity incentive plans. The Company does not have pending before the Commission any request for confidential treatment of information. Except for the issuance of the Securities contemplated by this Agreement, no event, liability, fact, circumstance, occurrence or development has occurred or exists or is reasonably expected to occur or exist with respect to the Company or its Subsidiaries or their respective businesses, prospects, properties, operations, assets or financial condition that would be required to be disclosed by the Company under applicable securities laws at the time this representation is made or deemed made that has not been publicly disclosed at least one (1) Trading Day prior to the date that this representation is made.
- (j) Litigation. There is no action, suit, inquiry, notice of violation, proceeding or investigation pending or, to the knowledge of the Company, threatened against or affecting the Company, any Subsidiary or any of their respective properties before or by any court, arbitrator, governmental or administrative agency or regulatory authority (federal, state, county, local or foreign) (collectively, an “Action”) except as disclosed in the General Disclosure Package. None of the Actions disclosed in the General Disclosure Package (i) adversely affects or challenges the legality, validity or enforceability of any of the Transaction Documents, the Shares or the Warrant Shares, or (ii) could, if there were an unfavorable decision, have or reasonably be expected to result in a Material Adverse Effect. Neither the Company nor any Subsidiary, nor any director or officer thereof, is or has been the subject of any Action involving a claim of violation of or liability under federal or state securities laws or a claim of breach of fiduciary duty, which could result in a Material Adverse Effect. There has not been, and to the knowledge of the Company, there is not pending or contemplated, any investigation by the Commission involving the Company or any current or former director or officer of the Company. The Commission has not issued any stop order or other order

suspending the effectiveness of any registration statement filed by the Company or any Subsidiary under the Exchange Act or the Securities Act.

- (k) Labor Relations. No labor dispute exists or, to the knowledge of the Company, is imminent with respect to any of the employees of the Company, which could reasonably be expected to result in a Material Adverse Effect. None of the Company's or its Subsidiaries' employees is a member of a union that relates to such employee's relationship with the Company or such Subsidiary, and neither the Company nor any of its Subsidiaries is a party to a collective bargaining agreement, and the Company and its Subsidiaries believe that their relationships with their employees are good. To the knowledge of the Company, no executive officer of the Company or any Subsidiary, is, or is now expected to be, in violation of any material term of any employment contract, confidentiality, disclosure or proprietary information agreement or non-competition agreement, or any other contract or agreement or any restrictive covenant in favor of any third party, and the continued employment of each such executive officer does not subject the Company or any of its Subsidiaries to any liability with respect to any of the foregoing matters. The Company and its Subsidiaries are in compliance with all applicable U.S. federal, state, local and foreign laws and regulations relating to employment and employment practices, terms and conditions of employment and wages and hours, except where the failure to be in compliance could not, individually or in the aggregate, reasonably be expected to have a Material Adverse Effect.
- (l) Compliance. Neither the Company nor any Subsidiary: (i) is in default under or in violation of (and no event has occurred that has not been waived that, with notice or lapse of time or both, would result in a default by the Company or any Subsidiary under), nor has the Company or any Subsidiary received notice of a claim that it is in default under or that it is in violation of, any indenture, loan or credit agreement or any other agreement or instrument to which it is a party or by which it or any of its properties is bound (whether or not such default or violation has been waived), (ii) is in violation of any judgment, decree or order of any court, arbitrator or other governmental authority or (iii) is or has been in violation of any statute, rule, ordinance or regulation of any governmental authority, including without limitation all foreign, federal, state and local laws relating to taxes, environmental protection, occupational health and safety, product quality and safety and employment and labor matters, except in each case of (i), (ii) and (iii) as could not have or reasonably be expected to result in a Material Adverse Effect.
- (m) Environmental Laws. The Company and its Subsidiaries (i) are in compliance with all federal, state, local and foreign laws relating to pollution or protection of human health or the environment (including ambient air, surface water, groundwater, land surface or subsurface strata), including laws relating to emissions, discharges, releases or threatened releases of chemicals, pollutants, contaminants, or toxic or hazardous substances or wastes (collectively, "Hazardous Materials") into the environment, or otherwise relating to the manufacture, processing, distribution, use, treatment, storage, disposal, transport or handling of Hazardous Materials, as well as all authorizations, codes, decrees, demands, or demand letters, injunctions, judgments, licenses, notices or notice letters, orders, permits, plans or regulations, issued, entered, promulgated or approved thereunder ("Environmental Laws"); (ii) have received all permits licenses or other approvals required of them under applicable Environmental Laws to conduct their respective businesses; and (iii) are in compliance with all terms and conditions of any such permit, license or approval where in each clause (i), (ii) and (iii), the failure to so comply could be reasonably expected to have, individually or in the aggregate, a Material Adverse Effect.
- (n) Regulatory Permits. The Company and the Subsidiaries possess all certificates, authorizations and permits issued by the appropriate federal, state, local or foreign regulatory authorities necessary to conduct their respective businesses as described in the General Disclosure Package, except where the failure to possess such certificates, authorizations or permits could not reasonably be expected to result in a Material Adverse Effect ("Material Permits"), and neither the Company nor any Subsidiary has received any notice of proceedings relating to the revocation or modification of any Material Permit.
- (o) Title to Assets. Except as disclosed in the General Disclosure Package, the Company and the Subsidiaries have good and marketable title in fee simple to all real property owned by them and good and marketable title in all personal property owned by them that is material to the business of the Company and the

Subsidiaries, in each case free and clear of all Liens, except for (i) Liens as do not materially affect the value of such property and do not materially interfere with the use made and proposed to be made of such property by the Company and the Subsidiaries and (ii) Liens for the payment of federal, state or other taxes, for which appropriate reserves have been made therefor in accordance with GAAP and, the payment of which is neither delinquent nor subject to penalties. Any real property and facilities held under lease by the Company and the Subsidiaries are held by them under valid, subsisting and enforceable leases with which the Company and the Subsidiaries are in compliance in all material respects.

- (p) Intellectual Property. The Company and the Subsidiaries have, or have rights to use, all patents, patent applications, trademarks, trademark applications, service marks, trade names, trade secrets, inventions, copyrights, licenses and other intellectual property rights and similar rights necessary or required for use in connection with their respective businesses as described in the General Disclosure Package, and which the failure to so have could have a Material Adverse Effect (collectively, the "Intellectual Property Rights"). None of, and neither the Company nor any Subsidiary has received a notice (written or otherwise) that any of, the Intellectual Property Rights has expired, terminated or been abandoned, or is expected to expire or terminate or be abandoned, within two (2) years from the date of this Agreement except as would not reasonably be expected to have a Material Adverse Effect. Neither the Company nor any Subsidiary has received, since the date of the latest audited financial statements included within the General Disclosure Package, a written notice of a claim or otherwise has any knowledge that the Intellectual Property Rights violate or infringe upon the rights of any Person, except as could not have or reasonably be expected to not have a Material Adverse Effect. To the actual knowledge of the Company, all such Intellectual Property Rights are enforceable and there is no existing infringement by another Person of any of the Intellectual Property Rights. The Company and its Subsidiaries have taken reasonable security measures to protect the secrecy, confidentiality and value of all of their intellectual properties, except where failure to do so could not, individually or in the aggregate, reasonably be expected to have a Material Adverse Effect.
- (q) Insurance. The Company and the Subsidiaries are insured by insurers of recognized financial responsibility against such losses and risks and in such amounts as are prudent and customary in the businesses in which the Company and the Subsidiaries are engaged, including, but not limited to, directors and officers insurance coverage in an amount deemed commercially reasonable. Neither the Company nor any Subsidiary has any reason to believe that it will not be able to renew its existing insurance coverage as and when such coverage expires or to obtain similar coverage from similar insurers as may be necessary to continue its business without a significant increase in cost.
- (r) Transactions with Affiliates and Employees. Except as set forth in the General Disclosure Package, none of the officers or directors of the Company or any Subsidiary and, to the knowledge of the Company, none of the employees of the Company or any Subsidiary is presently a party to any transaction with the Company or any Subsidiary (other than for services as employees, officers and directors), including any contract, agreement or other arrangement providing for the furnishing of services to or by, providing for rental of real or personal property to or from, providing for the borrowing of money from or lending of money to or otherwise requiring payments to or from any officer, director or such employee or, to the knowledge of the Company, any entity in which any officer, director, or any such employee has a substantial interest or is an officer, director, trustee, shareholder, member or partner, in each case in excess of \$120,000 other than for (i) payment of salary or consulting fees for services rendered, (ii) reimbursement for expenses incurred on behalf of the Company or a Subsidiary and (iii) other employee benefits, including share option agreements under any share option plan of the Company.
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- (s) Sarbanes-Oxley; Internal Accounting Controls. Except as set forth in the SEC Reports, the Company and the Subsidiaries are in material compliance with any and all applicable requirements of the Sarbanes-Oxley Act of 2002 that are effective as of the date hereof, and any and all applicable rules and regulations promulgated by the Commission thereunder that are effective as of the date hereof and as of the Closing Date. Except as set forth in the SEC Reports, the Company and the Subsidiaries maintain a system of internal accounting controls sufficient to provide reasonable assurance that: (i) transactions are executed in accordance with management's general or specific authorizations, (ii) transactions are recorded as necessary to permit preparation of financial statements in conformity with GAAP and to maintain asset accountability, (iii) access to assets is permitted only in accordance with management's general or specific authorization, and (iv) the recorded accountability for assets is compared with the existing assets at reasonable intervals and appropriate action is taken with respect to any differences. The Company and the Subsidiaries have established disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) for the Company and the Subsidiaries and designed such disclosure controls and procedures to ensure that information required to be disclosed by the Company in the reports it files or submits under the Exchange Act is recorded, processed, summarized and reported, within the time periods specified in the Commission's rules and forms. The Company's certifying officers have evaluated the effectiveness of the disclosure controls and procedures of the Company and the Subsidiaries as of the end of the period covered by the most recently filed Form 10-Q under the Exchange Act (such date, the "Evaluation Date"). The Company presented in its most recently filed Form 10-Q under the Exchange Act the conclusions of the certifying officers about the effectiveness of the disclosure controls and procedures based on their evaluations as of the Evaluation Date. Since the Evaluation Date, there have been no changes in the internal control over financial reporting (as such term is defined in the Exchange Act) that have materially affected, or is reasonably likely to materially affect, the internal control over financial reporting of the Company and its Subsidiaries.
- (t) Certain Fees. Except for fees payable to the Placement Agent, no brokerage or finder's fees or commissions are or will be payable by the Company or any Subsidiary to any broker, financial advisor or consultant, finder, placement agent, investment banker, bank or other Person with respect to the transactions contemplated by the Transaction Documents (for the avoidance of doubt, the foregoing shall not include any fees and/or commissions owed to the Depository). Other than for Persons engaged by any Purchaser, if any, the Purchasers shall have no obligation with respect to any fees or with respect to any claims made by or on behalf of other Persons for fees of a type contemplated in this Section that may be due in connection with the transactions contemplated by the Transaction Documents.
- (u) Investment Company. The Company is not, and is not an Affiliate of, and immediately after receipt of payment for the Securities, will not be or be an Affiliate of, an "investment company" within the meaning of the Investment Company Act of 1940, as amended. The Company shall conduct its business in a manner so that it will not become an "investment company" subject to registration under the Investment Company Act of 1940, as amended.
- (v) Registration Rights. No Person has any right to cause the Company to effect the registration under the Securities Act of any securities of the Company or any Subsidiary.
- (w) Listing and Maintenance Requirements. The shares of Common Stock are registered pursuant to Section 12(b) or 12(g) of the Exchange Act, and the Company has taken no action designed to, or which to its knowledge is likely to have the effect of, terminating the registration of the Common Stock under the Exchange Act nor has the Company received any notification that the Commission is contemplating terminating such registration. Except as disclosed in the General Disclosure Package, the Company has not, in the 12 months preceding the date hereof, received notice from any Trading Market on which the Common Stock are or have been listed or quoted to the effect that the Company is not in compliance with the listing or maintenance requirements of such Trading Market. The shares of Common Stock are currently eligible for electronic transfer through The Depository Trust Company or another established

clearing corporation and the Company is current in payment of the fees to The Depository Trust Company (or such other established clearing corporation) in connection with such electronic transfer.

- (x) Application of Takeover Protections. The Company and the Board of Directors have taken all necessary action, if any, in order to render inapplicable any control share acquisition, business combination, poison pill (including any distribution under a rights agreement) or other similar anti-takeover provision under the Company's articles of incorporation or the laws of its state of incorporation that is or could become applicable to the Purchasers as a result of the Purchasers and the Company fulfilling their obligations or exercising their rights under the Transaction Documents, including without limitation as a result of the Company's issuance of the Securities and the Purchasers' ownership of the Securities.
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- (y) Disclosure. Except with respect to the material terms and conditions of the transactions contemplated by the Transaction Documents, the Company confirms that it has not provided any of the Purchasers or their agents or counsel with any information that it believes constitutes or might constitute material, non-public information which is not otherwise disclosed in the Prospectus. The Company understands and confirms that the Purchasers will rely on the foregoing representation in effecting transactions in securities of the Company. All of the disclosure furnished by or on behalf of the Company to the Purchasers regarding the Company and its Subsidiaries, their respective businesses and the transactions contemplated hereby, including the Disclosure Schedules to this Agreement, is true and correct in all material respects and does not contain any untrue statement of a material fact or omit to state any material fact necessary in order to make the statements made therein, in the light of the circumstances under which they were made, not misleading. The press releases disseminated by the Company during the twelve months preceding the date of this Agreement taken as a whole do not contain any untrue statement of a material fact or omit to state a material fact required to be stated therein or necessary in order to make the statements therein, in the light of the circumstances under which they were made and when made, not misleading. The Company acknowledges and believes, to its best knowledge, that no Purchaser makes or has made any representations or warranties with respect to the transactions contemplated hereby other than those specifically set forth in Section 3.2 hereof.
- (z) No Integrated Offering. Assuming the accuracy of the Purchasers' representations and warranties set forth in Section 3.2, neither the Company, nor any of its Affiliates, nor any Person acting on its or their behalf has, directly or indirectly, made any offers or sales of any security or solicited any offers to buy any security, under circumstances that would cause this offering of the Securities to be integrated with prior offerings by the Company for purposes of any applicable stockholder approval provisions of any Trading Market on which any of the securities of the Company are listed or designated.
- (aa) Solvency. Except as disclosed in the General Disclosure Package, based on the consolidated financial condition of the Company as of the Closing Date, after giving effect to the receipt by the Company of the proceeds from the sale of the Securities hereunder, (i) the fair saleable value of the Company's assets exceeds the amount that will be required to be paid on or in respect of the Company's existing debts and other liabilities (including known contingent liabilities) as they mature, (ii) the Company's assets do not constitute unreasonably small capital to carry on its business as now conducted and as proposed to be conducted including its capital needs taking into account the particular capital requirements of the business conducted by the Company, consolidated and projected capital requirements and capital availability thereof, and (iii) the current cash flow of the Company, together with the proceeds the Company would receive, were it to liquidate all of its assets, after taking into account all anticipated uses of the cash, would be sufficient to pay all amounts on or in respect of its liabilities when such amounts are required to be paid. The Company does not intend to incur debts beyond its ability to pay such debts as they mature (taking into account the timing and amounts of cash to be payable on or in respect of its debt). The Company has no knowledge of any facts or circumstances which lead it to believe that it will file for reorganization or liquidation under the bankruptcy or reorganization laws of any jurisdiction within one year from the Closing Date. Schedule 3.1(aa) sets forth as of the date hereof all outstanding secured and unsecured Indebtedness of the Company or any Subsidiary, or for which the Company or any Subsidiary has commitments. For the purposes of this Agreement, "Indebtedness" means (x) any liabilities for borrowed money or amounts owed in excess of \$50,000 (other than trade accounts payable incurred in the ordinary course of business), (y) all guaranties, endorsements and other contingent obligations in respect of indebtedness of others, whether or not the same are or should be reflected in the Company's consolidated balance sheet (or the notes thereto), except guaranties by endorsement of negotiable instruments for deposit or collection or similar transactions in the ordinary course of business; and (z) the present value of any lease payments in excess of \$50,000 due under leases required to be capitalized in accordance with GAAP. Neither the Company nor any Subsidiary is in default with respect to any Indebtedness.

(bb) Tax Compliance. Except for matters that would not, individually or in the aggregate, have or reasonably be expected to result in a Material Adverse Effect, the Company and its Subsidiaries each (i) has made or filed all federal, state and local income and all foreign tax returns, reports and declarations required by any jurisdiction to which it is subject, (ii) has paid all taxes and other governmental assessments and charges, fines or penalties that are material in amount, shown or determined to be due on such returns, reports and declarations and (iii) has set aside on its financial statements provision reasonably adequate for the payment of all material tax liability of which has not been finally determined and all material taxes for periods subsequent to the periods to which such returns, reports or declarations apply. There are no unpaid taxes in any material amount claimed to be due by the taxing authority of any jurisdiction, and the officers of the Company or of any Subsidiary know of no basis for any such claim.

- (cc) Foreign Corrupt Practices. Neither the Company nor any Subsidiary, nor to the knowledge of the Company or any Subsidiary, any agent or other person acting on behalf of the Company or any Subsidiary, has (i) directly or indirectly, used any funds for unlawful contributions, gifts, entertainment or other unlawful expenses related to foreign or domestic political activity, (ii) made any unlawful payment to foreign or domestic government officials or employees or to any foreign or domestic political parties or campaigns from corporate funds, (iii) failed to disclose fully any contribution made by the Company or any Subsidiary (or made by any person acting on its behalf of which the Company is aware) which is in violation of law, or (iv) violated in any material respect any provision of FCPA.
- (dd) Accountants. The Company's independent registered public accounting firm is as set forth in the Prospectus. To the knowledge and belief of the Company, such accounting firm (i) is a registered public accounting firm as required by the Exchange Act and (ii) shall express its opinion with respect to the financial statements to be included in the Company's Annual Report for the fiscal year ending December 31, 2023.
- (ee) Acknowledgment Regarding Purchasers' Purchase of Securities. The Company acknowledges and agrees that each of the Purchasers is acting solely in the capacity of an arm's length purchaser with respect to the Transaction Documents and the transactions contemplated thereby. The Company further acknowledges that no Purchaser is acting as a financial advisor or fiduciary of the Company (or in any similar capacity) with respect to the Transaction Documents and the transactions contemplated thereby and any advice given by any Purchaser or any of their respective representatives or agents in connection with the Transaction Documents and the transactions contemplated thereby is merely incidental to the Purchasers' purchase of the Securities. The Company further represents to each Purchaser that the Company's decision to enter into this Agreement and the other Transaction Documents has been based solely on the independent evaluation of the transactions contemplated hereby by the Company and its representatives.
- (ff) Acknowledgment Regarding Purchaser's Trading Activity. Anything in this Agreement or elsewhere herein to the contrary notwithstanding (except for 4.12 hereof), it is understood and acknowledged by the Company that: (i) none of the Purchasers has been asked by the Company to agree, nor has any Purchaser agreed, to desist from purchasing or selling, long and/or short, securities of the Company, or "derivative" securities based on securities issued by the Company or to hold the Securities for any specified term; (ii) past or future open market or other transactions by any Purchaser, specifically including, without limitation, Short Sales or "derivative" transactions, before or after the closing of this or future private placement transactions, may negatively impact the market price of the Company's publicly-traded securities; (iii) any Purchaser, and counter-parties in "derivative" transactions to which any such Purchaser is a party, directly or indirectly, presently may have a "short" position in the shares of Common Stock, and (iv) each Purchaser shall not be deemed to have any affiliation with or control over any arm's length counter-party in any "derivative" transaction. The Company further understands and acknowledges that (y) one or more Purchasers may engage in hedging activities at various times during the period that the Securities are outstanding, and (z) such hedging activities (if any) could reduce the value of the existing stockholders' equity interests in the Company at and after the time that the hedging activities are being conducted. The Company acknowledges that such aforementioned hedging activities do not constitute a breach of any of the Transaction Documents.
- (gg) Regulation M Compliance. The Company has not, and to its knowledge no one acting on its behalf has, (i) taken, directly or indirectly, any action designed to cause or to result in the stabilization or manipulation of the price of any security of the Company to facilitate the sale or resale of any of the shares of Common Stock, (ii) sold, bid for, purchased, or paid any compensation for soliciting purchases of, any of the shares of Common Stock, or (iii) paid or agreed to pay to any Person any compensation for soliciting another to purchase any other securities of the Company, other than, in the case of clauses (ii) and (iii), compensation paid to the Placement Agent in connection with the placement of the shares of Common Stock.

- (hh) Stock Option Plans. Each stock option granted by the Company under the Company's stock incentive plans was granted (i) in accordance with the terms of such plan and (ii) with an exercise price at least equal to the fair market value of the Common Stock on the date such stock option would be considered granted under GAAP and applicable law. No stock option granted under the Company's stock option plan has been backdated.
- (ii) Cybersecurity. Except as would not, individually or in the aggregate, have a Material Adverse Effect, (i) the Company and the Subsidiaries are presently in compliance with all applicable laws or statutes and all judgments, orders, rules and regulations of any court or arbitrator or governmental or regulatory authority, internal policies and contractual obligations relating to the privacy and security of the Company's or any Subsidiary's information technology and computer systems, networks, hardware, software, data (including the data of its respective customers, employees, suppliers, vendors and any third party data maintained by or on behalf of it), equipment or technology (collectively, "IT Systems and Data") and to the protection of such IT Systems and Data from unauthorized use, access, misappropriation or modification; (ii) the Company and the Subsidiaries have implemented and maintained commercially reasonable safeguards to maintain and protect its material confidential information and the integrity, continuous operation, redundancy and security of all IT Systems and Data; and (iii) the Company and the Subsidiaries have implemented backup and disaster recovery technology consistent with commercially reasonable industry standards and practices.
- (jj) FDA. As to each product subject to the jurisdiction of the U.S. Food and Drug Administration ("FDA") under the Federal Food, Drug and Cosmetic Act, as amended, and the regulations thereunder ("FDCA") that is manufactured, packaged, labeled, tested, distributed, sold, and/or marketed by the Company or any of its Subsidiaries (each such product, a "Pharmaceutical Product"), such Pharmaceutical Product is being manufactured, packaged, labeled, tested, distributed, sold and/or marketed by the Company in compliance with all applicable requirements under FDCA and similar laws, rules and regulations relating to registration, investigational use, premarket clearance, licensure, or application approval, good manufacturing practices, good laboratory practices, good clinical practices, product listing, quotas, labeling, advertising, record keeping and filing of reports, except where the failure to be in compliance would not have a Material Adverse Effect. There is no pending, completed or, to the Company's knowledge, threatened, action (including any lawsuit, arbitration, or legal or administrative or regulatory proceeding, charge, complaint, or investigation) against the Company or any of its Subsidiaries, and none of the Company or any of its Subsidiaries has received any notice, warning letter or other communication from the FDA or any other governmental entity, which (i) contests the premarket clearance, licensure, registration, or approval of, the uses of, the distribution of, the manufacturing or packaging of, the testing of, the sale of, or the labeling and promotion of any Pharmaceutical Product, (ii) withdraws its approval of, requests the recall, suspension, or seizure of, or withdraws or orders the withdrawal of advertising or sales promotional materials relating to, any Pharmaceutical Product, (iii) imposes a clinical hold on any clinical investigation by the Company or any of its Subsidiaries, (iv) enjoins production at any facility of the Company or any of its Subsidiaries, (v) enters or proposes to enter into a consent decree of permanent injunction with the Company or any of its Subsidiaries, or (vi) otherwise alleges any violation of any laws, rules or regulations by the Company or any of its Subsidiaries, and which, either individually or in the aggregate, would have a Material Adverse Effect. The properties, business and operations of the Company have been and are being conducted in all material respects in accordance with all applicable laws, rules and regulations of the FDA. The Company has not been informed by the FDA that the FDA will prohibit the marketing, sale, license or use in the United States of any product proposed to be developed, produced or marketed by the Company nor has the FDA expressed any concern as to approving or clearing for marketing any product being developed or proposed to be developed by the Company.
- (kk) Office of Foreign Assets Control. Neither the Company nor any Subsidiary nor, to the Company's knowledge, any director, officer, agent, employee or affiliate of the Company or any Subsidiary is currently

subject to any U.S. sanctions administered by the Office of Foreign Assets Control of the U.S. Treasury Department ("OFAC").

- (II) Money Laundering. The operations of the Company and its Subsidiaries are and have been conducted at all times in compliance with applicable financial record-keeping and reporting requirements of the Currency and Foreign Transactions Reporting Act of 1970, as amended, applicable money laundering statutes and applicable rules and regulations thereunder (collectively, the “Money Laundering Laws”), and no action, suit or proceeding by or before any court or governmental agency, authority or body or any arbitrator involving the Company or any Subsidiary with respect to the Money Laundering Laws is pending or, to the knowledge of the Company or any Subsidiary, threatened.

3.2 Representations and Warranties of the Purchasers. Each Purchaser, for itself and for no other Purchaser, hereby represents and warrants as of the date hereof and as of the Closing Date to the Company as follows (unless as of a specific date therein, in which case they shall be accurate as of such date):

- (a) Organization; Authority. Such Purchaser is either an individual or an entity duly incorporated or formed, validly existing and in good standing under the laws of the jurisdiction of its incorporation or formation with full right, corporate, partnership limited liability company or similar power and authority to enter into and to consummate the transactions contemplated by the Transaction Documents and otherwise to carry out its obligations hereunder and thereunder. The execution and delivery of the Transaction Documents and performance by such Purchaser of the transactions contemplated by the Transaction Documents have been duly authorized by all necessary corporate, partnership, limited liability company or similar action, as applicable, on the part of such Purchaser. Each Transaction Document to which it is a party has been duly executed by such Purchaser, and when delivered by such Purchaser in accordance with the terms hereof, will constitute the valid and legally binding obligation of such Purchaser, enforceable against it in accordance with its terms, except: (i) as limited by general equitable principles and applicable bankruptcy, insolvency, reorganization, moratorium and other laws of general application affecting enforcement of creditors’ rights generally, (ii) as limited by laws relating to the availability of specific performance, injunctive relief or other equitable remedies and (iii) insofar as indemnification and contribution provisions may be limited by applicable law.
- (b) Understandings or Arrangements. Such Purchaser is acquiring the Securities as principal for its own account and has no direct or indirect arrangement or understandings with any other persons to distribute or regarding the distribution of such Securities (this representation and warranty not limiting such Purchaser’s right to sell the Securities pursuant to the Registration Statement or otherwise in compliance with applicable federal and state securities laws).
- (c) Experience of Such Purchaser. Such Purchaser, either alone or together with its representatives, has such knowledge, sophistication and experience in business and financial matters so as to be capable of evaluating the merits and risks of the prospective investment in the Securities, and has so evaluated the merits and risks of such investment. Such Purchaser is able to bear the economic risk of an investment in the Securities and, at the present time, is able to afford a complete loss of such investment.
- (d) Certain Transactions and Confidentiality. Other than consummating the transactions contemplated hereunder, such Purchaser has not, nor has any Person acting on behalf of or pursuant to any understanding with such Purchaser, directly or indirectly executed any purchases or sales, including Short Sales, of the securities of the Company during the period commencing as of the time that such Purchaser first received a term sheet (written or oral) from the Company or any other Person representing the Company setting forth the material terms, which terms include definitive pricing terms, of the transactions contemplated hereunder and ending immediately prior to the execution hereof. Notwithstanding the foregoing, in the case of a Purchaser that is a multi-managed investment vehicle whereby separate portfolio managers manage separate portions of such Purchaser’s assets and the portfolio managers have no direct knowledge of the investment decisions made by the portfolio managers managing other portions of such Purchaser’s assets, the representation set forth above shall only apply with respect to the portion of assets managed by the portfolio manager that made the investment decision to purchase the Securities covered by this Agreement.

Other than to other Persons party to this Agreement or to such Purchaser's representatives, including, without limitation, its officers, directors, partners, legal and other advisors, employees, agents and Affiliates, such Purchaser has maintained the confidentiality of all disclosures made to it in connection with this transaction (including the existence and terms of this transaction). Notwithstanding the foregoing, for the avoidance of doubt, nothing contained herein shall constitute a representation or warranty, or preclude any actions, with respect to locating or borrowing shares order to effect Short Sales or similar transactions in the future.

- (e) Independent Advice. Each Purchaser understands that nothing in this Agreement or any other materials presented by or on behalf of the Company to the Purchaser in connection with the purchase of the Securities constitutes legal, tax or investment advice.

The Company acknowledges and agrees that the representations contained in this Section 3.2 shall not modify, amend or affect such Purchaser's right to rely on the Company's representations and warranties contained in this Agreement or any representations and warranties contained in any other Transaction Document or any other document or instrument executed and/or delivered in connection with this Agreement or the consummation of the transactions contemplated hereby. Notwithstanding the foregoing, for the avoidance of doubt, nothing contained herein shall constitute a representation or warranty, or preclude any actions, except as set forth in this Agreement, with respect to locating or borrowing shares in order to effect Short Sales or similar transactions in the future.

Section 4.

OTHER AGREEMENTS OF THE PARTIES

4.1 Legends. The shares of Common Stock and the Warrant Shares shall be issued free of legends. If at any time following the date hereof the Registration Statement is not effective or is not otherwise available for the sale of the shares of Common Stock, the Warrants or the Warrant Shares, the Company shall immediately notify the holders of the Warrants in writing that such registration statement is not then effective and thereafter shall promptly notify such holders when the registration statement is effective again and available for the sale of the Shares, the Warrants or the Warrant Shares (it being understood and agreed that the foregoing shall not limit the ability of the Company to issue, or any Purchaser to sell, any of the Shares, the Warrants or the Warrant Shares in compliance with applicable federal and state securities laws). The Company shall use commercially reasonable best efforts to keep a registration statement (including the Registration Statement) registering the issuance of the Warrant Shares effective during the term of the Warrants.

4.2 Furnishing of Information; Public Information. Until the earliest of the time that (i) no Purchaser owns Securities, or (ii) the Common Warrants have expired, the Company covenants to maintain the registration of the Common Stock under Section 12(b) or 12(g) of the Exchange Act and to timely file (or obtain extensions in respect thereof and file within the applicable grace period) all reports required to be filed by the Company after the date hereof pursuant to the Exchange Act, even if the Company is not then subject to the reporting requirements of the Exchange Act, (in each case on or after the date as of which the Purchasers may sell all of their Securities without restriction or limitation pursuant to Rule 144) except in the event that the Company consummates: (a) any transaction or series of related transactions as a result of which any Person (together with its Affiliates) acquires then outstanding securities of the Company representing more than fifty percent (50%) of the voting control of the Company; (b) a merger or reorganization of the Company with one or more other entities in which the Company is not the surviving entity; or (c) a sale of all or substantially all of the assets of the Company, where the consummation of such transaction results in the Company no longer being subject to the reporting requirements of the Exchange Act.

4.3 Integration. The Company shall not sell, offer for sale or solicit offers to buy or otherwise negotiate in respect of any security (as defined in Section 2 of the Securities Act) that would be integrated with the offer or sale of the Securities for purposes of the rules and regulations of any Trading Market such that it would require shareholder approval prior to the closing of such other transaction unless shareholder approval is obtained before the closing of such subsequent transaction.

4.4 Securities Laws Disclosure; Publicity. The Company shall (a) by the Disclosure Time, issue a press release disclosing the material terms of the transactions contemplated hereby and other material non-public information delivered to any of the Purchasers by the Company or any of its Subsidiaries, or any of their respective officers, directors, employees or agents, and (b) file a Current Report on Form 8-K, including the Transaction Documents as exhibits thereto, with the Commission within the time required by the Exchange Act. From and after the issuance of such press release, the Company represents to the Purchasers that it shall have publicly disclosed all material, non-public information delivered to any of the Purchasers by the Company or any of its Subsidiaries, or any of their respective officers, directors, employees or agents. In addition, effective upon the issuance of such press release, the Company acknowledges and agrees that any and all confidentiality or similar obligations under any agreement, whether written or oral, between the Company, any of its Subsidiaries or any of their respective officers, directors, agents, employees or Affiliates on the one hand, and any of the Purchasers or any of their Affiliates on the other hand, shall terminate. The Company and each Purchaser shall consult with each other in issuing any other press releases with respect to the transactions contemplated hereby, and neither the Company nor any Purchaser shall issue any such press release nor otherwise make any such public statement without the prior consent of the Company, with respect to any press release of any Purchaser, or without the prior consent of each Purchaser, with respect to any press release of the Company, which consent shall not unreasonably be withheld or delayed, except if such disclosure is required by law, in which case the disclosing party shall promptly provide the other party with prior notice of such public statement or communication. Notwithstanding the foregoing, the Company shall not publicly disclose the name of any Purchaser, or include the name of any Purchaser in any filing with the Commission or any regulatory agency or Trading Market, without the prior written consent of such Purchaser, except (a) as required by federal securities law in connection with the filing of final Transaction Documents with the Commission, and (b) to the extent such disclosure is required by law or Trading Market regulations, in which case the Company shall provide the Purchasers with prior notice of such disclosure permitted under this clause (b).

4.5 Stockholder Rights Plan. No claim will be made or enforced by the Company or, with the consent of the Company, any other Person, that any Purchaser is an "Acquiring Person" under any control share acquisition, business combination, poison pill (including any distribution under a rights agreement) or similar anti-takeover plan or arrangement in effect or hereafter adopted by the Company, or that any Purchaser could be deemed to trigger the provisions of any such plan or arrangement, by virtue of receiving Securities under the Transaction Documents or under any other agreement between the Company and the Purchasers.

4.6 Non-Public Information. Except with respect to the material terms and conditions of the transactions contemplated by the Transaction Documents, which shall be disclosed pursuant to Section 4.4, the Company covenants and agrees that neither it, nor any other Person acting on its behalf will provide any Purchaser or its agents or counsel with any information that the Company reasonably believes constitutes material non-public information, unless prior thereto such Purchaser shall have entered into a written agreement with the Company regarding the confidentiality and use of such information. The Company understands and confirms that each Purchaser shall be relying on the foregoing covenant in effecting transactions in securities of the Company. To the extent that the Company delivers any material, non-public information to a Purchaser without such Purchaser's consent, the Company hereby covenants and agrees that such Purchaser shall not have any duty of confidentiality to the Company, any of its Subsidiaries, or any of their respective officers, directors, agents, employees or Affiliates, or a duty to the Company, and of its Subsidiaries or any of their respective officers, directors, agents, employees or Affiliates not to trade on the basis of, such material, non-public information, provided that the Purchaser shall remain subject to applicable law. To the extent that any notice provided pursuant to any Transaction Document constitutes, or contains, material, non-public information regarding the Company or any Subsidiaries, the Company shall simultaneously file such notice with the Commission pursuant to a Current Report on Form 8-K. The Company understands and confirms that each Purchaser shall be relying on the foregoing covenant in effecting transactions in securities of the Company.

4.7 Use of Proceeds. Although the Company intends to use the net proceeds from the sale of the Securities hereunder for general working capital purposes and potential acquisitions], the Company has discretion in the application of the net proceeds of the Offering.

4.8 Indemnification of Purchasers. Subject to the provisions of this Section 4.8, the Company will indemnify and hold each Purchaser and its directors, officers, shareholders, members, partners, employees and agents (and any other Persons with a functionally equivalent role of a Person holding such titles notwithstanding a lack of such title or any other title), each Person who controls such Purchaser (within the meaning of Section 15 of the Securities Act and Section 20 of the Exchange Act), and the directors, officers, shareholders, agents, members, partners or employees (and any other Persons with a functionally equivalent role of a Person holding such titles notwithstanding a lack of such title or any other title) of such controlling persons (each, a "Purchaser Party") harmless from any and all losses, liabilities, obligations, claims, contingencies, damages, costs and expenses, including all judgments, amounts paid in settlements, court costs and reasonable attorneys' fees and costs of investigation that any such Purchaser Party may suffer or incur as a result of or relating to (a) any breach of any of the representations, warranties, covenants or agreements made by the Company in this Agreement or in the other Transaction Documents or (b) any action instituted against the Purchaser Parties in any capacity (including a Purchaser Party's status as an investor), or any of them or their respective Affiliates, any stockholder of the Company who is not an Affiliate of such Purchaser Party, arising out of or relating to any of the transactions contemplated by the Transaction Documents (unless such action is based upon a material breach of such Purchaser Party's representations, warranties, covenants or agreements made by such Purchaser Party in any Transaction Document or any conduct by a Purchaser Party which is finally judicially determined to constitute fraud, gross negligence or willful misconduct). If any action shall be brought against any Purchaser Party in respect of which indemnity may be sought pursuant to this Agreement, such Purchaser Party shall promptly notify counsel of its own choosing reasonably acceptable to the Purchaser Party. Any Purchaser Party shall have the right to employ separate counsel in any such action and participate in the defense thereof, but the fees and expenses of such counsel shall be at the expense of such Purchaser Party except to the extent that (i) the employment thereof has been specifically authorized by the Company in writing, (ii) the Company has failed after a reasonable period of time to assume such defense and to employ counsel or (iii) in such action there is, in the reasonable opinion of counsel to the applicable Purchaser Party (which may be internal counsel), a material conflict on any material issue between the position of the Company and the position of such Purchaser Party, in which case the Company shall be responsible for the reasonable fees and expenses of no more than one such separate counsel. The Company will not be liable to any Purchaser Party under this Agreement for any settlement by a Purchaser Party effected without the Company's prior written consent, which shall not be unreasonably withheld or delayed, or the extent that a loss, claim, damage, or liability is attributable to any Purchaser Party's breach of any of the representations, warranties, covenants or agreements made by such Purchaser Party in this Agreement or in the other Transaction Documents. The indemnification and other payment obligations required by this Section 4.8 shall be made by periodic payments of the amount thereof during the course of the investigation, defense, collection, enforcement or action, as and when bills are received or are incurred; provided, that if any Purchaser Party is finally judicially determined not to be entitled to indemnification or payment under this Section 4.8, such Purchaser Party shall promptly reimburse the Company for any payments that are advanced under this sentence. The indemnity agreements contained herein shall be in addition to any cause of action or similar right of any Purchaser Party against the Company or others and any liabilities the Company may be subject to pursuant to law.

4.9 Listing of Shares. The Company hereby agrees to use commercially reasonable best efforts to maintain the listing or quotation of the shares of Common Stock on each Trading Market on which each is currently listed, and concurrently with the Closing, the Company shall apply to list or quote all of the shares of Common Stock on such Trading Markets and promptly secure the listing of all of the shares of Common Stock on such Trading Markets. The Company further agrees, if the Company applies to have the Common Stock traded on any other Trading Market, it will then include in such application all of the shares of Common Stock and Warrant Shares, and will take such other action as is necessary to cause all of the shares of Common Stock and Warrant Shares to be listed or quoted on such other Trading Market as promptly as possible. The Company will then take all action reasonably necessary to continue the listing and trading of the Common Stock on a Trading Market and will comply in all material respects with the Company's reporting, filing and other obligations under the bylaws or rules of the Trading Market. The Company agrees to use commercially reasonable efforts to maintain the eligibility of the for electronic transfer through the Depository Trust Company or another established clearing corporation, including, without limitation, by timely payment of fees to the Depository Trust Company or such other established clearing corporation in connection with such electronic transfer. Notwithstanding anything to the contrary contained in this Agreement, none of the foregoing shall apply in the event that the Company consummates (a) any transaction or series of related transactions as a result of which any Person (together with its

Affiliates) acquires then outstanding securities of the Company representing more than fifty percent (50%) of the voting control of the Company; (b) a merger or reorganization of the Company with one or more other entities in which the Company is not the surviving entity; or (c) a sale of all or substantially all of the assets of the Company, where the consummation of such transaction results in the Company no longer being subject to the reporting requirements of the Exchange Act

4.10 Subsequent Equity Sales.

- (a) From the date hereof until ninety (90) days after the Closing Date, neither the Company nor any Subsidiary shall (i) issue, enter into any agreement to issue or announce the issuance or proposed issuance of any shares of Common Stock or Common Stock Equivalents or (ii) file any registration statement or amendment or supplement thereto, other than the Prospectus Supplement or filing a registration statement on Form S-8 in connection with any employee benefit plan.
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- (b) From the date hereof until the six (6) month anniversary of the Closing Date, the Company shall be prohibited from effecting or entering into an agreement to effect any issuance by the Company or any of its Subsidiaries of shares of Common Stock or Common Stock Equivalents (or a combination of units thereof) involving a Variable Rate Transaction. "Variable Rate Transaction" means a transaction in which the Company (i) issues or sells any debt or equity securities that are convertible into, exchangeable or exercisable for, or include the right to receive additional shares of Common Stock either (A) at a conversion price, exercise price or exchange rate or other price that is based upon and/or varies with the trading prices of or quotations for the Common Stock at any time after the initial issuance of such debt or equity securities, or (B) with a conversion, exercise or exchange price that is subject to being reset at some future date after the initial issuance of such debt or equity security or upon the occurrence of specified or contingent events directly or indirectly related to the business of the Company or the market for shares of Common Stock or (ii) enters into, or effects a transaction under, any agreement, including, but not limited to, an equity line of credit or an "at-the-market offering", whereby the Company may issue securities at a future determined price regardless of whether shares pursuant to such agreement have actually been issued and regardless of whether such agreement is subsequently canceled. Any Purchaser shall be entitled to obtain injunctive relief against the Company to preclude any such issuance, which remedy shall be in addition to any right to collect damages.
- (c) Notwithstanding the foregoing, this Section 4.10 shall not apply (1) in respect of an Exempt Issuance, except that no Variable Rate Transaction shall be an Exempt Issuance (2) following the forty-five (45) day anniversary of the Closing Date, to the filing of a prospectus supplement with respect to sales under the at-the-market offering in which the Placement Agent is a sales agent, pursuant to the Amended and Restated At Market Issuance Sales Agreement, dated February 9, 2021, between the Company and the Placement Agent, as amended by Amendment No. 1 thereto, dated May 3, 2021, and Amendment No. 2 thereto, dated May 2, 2024 (the "Sales Agreement") and (3) following the seventy-five (75) day anniversary of the Closing Date, to the sale and issuance of securities in an at-the-market offering pursuant to the Sales Agreement.

4.11 Equal Treatment of Purchasers. No consideration (including any modification of the Transaction Documents) shall be offered or paid to any Person to amend or consent to a waiver or modification of any provision of the Transaction Documents unless the same consideration is also offered to all of the parties to the Transaction Documents. For clarification purposes, this provision constitutes a separate right granted to each Purchaser by the Company and negotiated separately by each Purchaser, and is intended for the Company to treat the Purchasers as a class and shall not in any way be construed as the Purchasers acting in concert or as a group with respect to the purchase, disposition or voting of the shares of Common Stock or otherwise.

4.12 Certain Transactions and Confidentiality. Each Purchaser, severally and not jointly with the other Purchasers, covenants that neither it nor any Affiliate acting on its behalf or pursuant to any understanding with it will execute any purchases or sales, including Short Sales of any of the Company's securities during the period commencing with the execution of this Agreement and ending at such time that the transactions contemplated by this Agreement are first publicly announced pursuant to the initial press release as described in Section 4.4. Each Purchaser, severally and not jointly with the other Purchasers, covenants that until such time as the transactions contemplated by this Agreement are publicly disclosed by the Company pursuant to the initial press release as described in Section 4.4, such Purchaser will maintain the confidentiality of the existence and terms of this transaction. Notwithstanding the foregoing, and notwithstanding anything contained in this Agreement to the contrary, the Company expressly acknowledges and agrees that (i) no Purchaser makes any representation, warranty or covenant hereby that it will not engage in effecting transactions in any securities of the Company after the time that the transactions contemplated by this Agreement are first publicly announced pursuant to the initial press release as described in Section 4.4, (ii) no Purchaser shall be restricted or prohibited from effecting any transactions in any securities of the Company in accordance with applicable securities laws from and after the time that the transactions contemplated by this Agreement are first publicly announced pursuant to the initial press release as described in Section 4.4 and (iii) no Purchaser shall have any duty of confidentiality or duty not to trade in the securities of the Company to the Company or its Subsidiaries after the issuance

of the initial press release as described in Section 4.4. Notwithstanding the foregoing, in the case of a Purchaser that is a multi-managed investment vehicle whereby separate portfolio managers manage separate portions of such Purchaser's assets and the portfolio managers have no direct knowledge of the investment decisions made by the portfolio managers managing other portions of such Purchaser's assets, the covenant set forth above shall only apply with respect to the portion of assets managed by the portfolio manager that made the investment decision to purchase the Securities covered by this Agreement.

4.13 Exercise Procedures. The form of Notice of Exercise included in the Warrants set forth the totality of the procedures required of the Purchasers in order to exercise the Warrants. No additional legal opinion, other information or instructions shall be required of the Purchasers to exercise their Warrants. Without limiting the preceding sentences, no ink-original Notice of Exercise shall be required, nor shall any medallion guarantee (or other type of guarantee or notarization) of any Notice of Exercise form be required in order to exercise the Warrants. The Company shall honor exercises of the Warrants and shall deliver shares of Common Stock and/or Warrant Shares in accordance with the terms, conditions and time periods set forth in the Transaction Documents.

4.14 Reservations of Shares. As of the date hereof, the Company has reserved and the Company shall continue to reserve and keep available at all times, free of preemptive rights, a sufficient number of shares of Common Stock for the purpose of enabling the Company to issue shares of Common Stock pursuant to this Agreement and Warrant Shares pursuant to any exercise of the Warrants.

4.15 Lock-Up Agreements. The Company shall not amend, modify, waive or terminate any provision of any of the Lock-Up Agreements without the prior written consent of the Placement Agent, except to extend the term of the lock-up period, and shall enforce the provisions of each Lock-Up Agreement in accordance with its terms. If any party to a Lock-Up Agreement breaches any provision of a Lock-Up Agreement, the Company shall promptly use its best efforts to seek specific performance of the terms of such Lock-Up Agreement.

Section 5. MISCELLANEOUS

5.1 Termination. This Agreement may be terminated by any Purchaser, as to such Purchaser's obligations hereunder only and without any effect whatsoever on the obligations between the Company and the other Purchasers, by written notice to the other parties, if the Closing has not been consummated on or before the fifth (5th) Trading Day following the date hereof; provided, however, that no such termination will affect the right of any party to sue for any breach by any other party (or parties).

5.2 Fees and Expenses. Except as expressly set forth in the Transaction Documents to the contrary, each party shall pay the fees and expenses of its advisers, counsel, accountants and other experts, if any, and all other expenses incurred by such party incident to the negotiation, preparation, execution, delivery and performance of this Agreement. The Company shall pay all Depositary Fees (including, without limitation, any fees required for same-day processing of any instruction letter delivered by the Company and any exercise notice delivered by a Purchaser), stamp taxes and other taxes and duties levied in connection with the delivery of any Securities to the Purchasers.

5.3 Entire Agreement. The Transaction Documents, together with the exhibits and schedules thereto, the Preliminary Prospectus and the Prospectus, contain the entire understanding of the parties with respect to the subject matter hereof and thereof and supersede all prior agreements and understandings, oral or written, with respect to such matters, which the parties acknowledge have been merged into such documents, exhibits and schedules.

5.4 Notices. Any and all notices or other communications or deliveries required or permitted to be provided hereunder shall be in writing and shall be deemed given and effective on the earliest of: (a) the time of transmission, if such notice or communication is delivered via facsimile at the facsimile number or email attachment at the email address as set forth on the signature pages attached hereto at or prior to 5:30 p.m. (New York City time) on a Trading Day, (b) the next Trading Day after the time of transmission, if such notice or communication is delivered via facsimile at the facsimile number or email attachment at the email address as set forth on the signature pages attached hereto on a day that is not a Trading Day or later than 5:30 p.m. (New York City time) on any Trading Day, (c) the second (2nd) Trading Day following the date of mailing, if sent by U.S. nationally recognized overnight courier service or (d) upon actual receipt by the party to whom such notice is required to be given. The address for such notices and communications shall be as set forth on the signature pages attached hereto. To the extent that any notice provided pursuant to any Transaction Document constitutes, or contains, material, non-public information regarding the Company or any Subsidiaries, the Company shall simultaneously file such notice with the Commission pursuant to a Current Report on Form 8-K.

5.5 Amendments; Waivers. No provision of this Agreement may be waived, modified, supplemented or amended except in a written instrument signed, in the case of an amendment, by the Company and the Purchasers who execute this Agreement and who purchased at least 50.1% in interest of the sum of (i) the Shares and (ii) the Pre-Funded Warrant Shares initially issuable upon exercise of the Pre-Funded Warrants based on the initial Subscription Amounts hereunder (or prior to the Closing, the Company and each Purchaser), or, in the case of a waiver, by the party against whom enforcement of any such waived provision is sought; provided, that if any amendment, modification or waiver disproportionately and adversely impacts a Purchaser (or multiple Purchasers), the consent of such disproportionately impacted Purchaser (or at least 50.1% in interest of such multiple Purchasers) shall also be required. No waiver of any default with respect to any provision, condition or requirement of this Agreement shall be deemed to be a continuing waiver in the future or a waiver of any subsequent default or a waiver of any other provision, condition or requirement hereof, nor shall any delay or omission of any party to exercise any right hereunder in any manner impair the exercise of any such right. Any proposed amendment or waiver that disproportionately, materially and adversely affects the rights and obligations of any Purchaser relative to the comparable rights and obligations of the other Purchasers shall require the prior written consent of such adversely affected Purchaser. Any amendment effected in accordance with this Section 5.5 shall be binding upon each Purchaser and holder of Securities and the Company.

5.6 Headings. The headings herein are for convenience only, do not constitute a part of this Agreement and shall not be deemed to limit or affect any of the provisions hereof.

5.7 Successors and Assigns. This Agreement shall be binding upon and inure to the benefit of the parties and their successors and permitted assigns. The Company may not assign this Agreement or any rights or obligations hereunder without the prior written consent of each Purchaser (other than by merger). Any Purchaser may assign any or all of its rights under this Agreement to any Person to whom such Purchaser assigns or transfers any Securities, provided that such transferee agrees in writing to be bound, with respect to the transferred Securities, by the provisions of the Transaction Documents that apply to the "Purchasers."

5.8 No Third-Party Beneficiaries. The Placement Agent shall be the third-party beneficiary of the representations and warranties of the Company in Section 3.1 and the representations and warranties of the Purchasers in Section 3.2. This Agreement is intended for the benefit of the parties hereto and their respective successors and permitted assigns and is not for the benefit of, nor may any provision hereof be enforced by, any other Person, except as otherwise set forth in Section 4.8 and this Section 5.8.

5.9 Governing Law. All questions concerning the construction, validity, enforcement and interpretation of the Transaction Documents shall be governed by and construed and enforced in accordance with the internal laws of the State of New York, without regard to the principles of conflicts of law thereof. Each party agrees that all legal proceedings concerning the interpretations, enforcement and defense of the transactions contemplated by this Agreement and any other Transaction Documents (whether brought against a party hereto or its respective affiliates, directors, officers, shareholders, partners, members, employees or agents) shall be commenced exclusively in the state and federal courts sitting in the state of New York. Each party hereby irrevocably submits to the exclusive jurisdiction of the state and federal courts sitting in the state of New York for the adjudication of any dispute hereunder or in connection herewith or with any transaction contemplated hereby or discussed herein (including with respect to the enforcement of any of the Transaction Documents), and hereby irrevocably waives, and agrees not to assert in any suit, action or proceeding, any claim that it is not personally subject to the jurisdiction of any such court, that such suit, action or proceeding is improper or is an inconvenient venue for such proceeding. Each party hereby irrevocably waives personal service of process and consents to process being served in any such suit, action or proceeding by mailing a copy thereof via registered or certified mail or overnight delivery (with evidence of delivery) to such party at the address in effect for notices to it under this Agreement and agrees that such service shall constitute good and sufficient service of process and notice thereof. Nothing contained herein shall be deemed to limit in any way any right to serve process in any other manner permitted by law. If either party shall commence an action, suit or proceeding to enforce any provisions of the Transaction Documents, then, in addition to the obligations of the Company under Section 4.8, the prevailing party in such action, suit or proceeding shall be reimbursed by the other party for its reasonable attorneys' fees and other costs and expenses incurred with the investigation, preparation and prosecution of such action or proceeding.

5.10 Survival. The representations and warranties contained herein shall survive the Closing and the delivery of the Securities for the applicable statute of limitations.

5.11 Execution. This Agreement may be executed in two or more counterparts, all of which when taken together shall be considered one and the same agreement and shall become effective when counterparts have been signed by each party and delivered to each other party, it being understood that the parties need not sign the same counterpart. In the event that any signature is delivered by facsimile transmission or by e-mail delivery of a “.pdf” format data file, such signature shall create a valid and binding obligation of the party executing (or on whose behalf such signature is executed) with the same force and effect as if such facsimile or “.pdf” signature page was an original thereof.

5.12 Severability. If any term, provision, covenant or restriction of this Agreement is held by a court of competent jurisdiction to be invalid, illegal, void or unenforceable, the remainder of the terms, provisions, covenants and restrictions set forth herein shall remain in full force and effect and shall in no way be affected, impaired or invalidated, and the parties hereto shall use their commercially reasonable efforts to find and employ an alternative means to achieve the same or substantially the same result as that contemplated by such term, provision, covenant or restriction. It is hereby stipulated and declared to be the intention of the parties that they would have executed the remaining terms, provisions, covenants and restrictions without including any of such that may be hereafter declared invalid, illegal, void or unenforceable.

5.13 Rescission and Withdrawal Right. Notwithstanding anything to the contrary contained in (and without limiting any similar provisions of) any of the other Transaction Documents, whenever any Purchaser exercises a right, election, demand or option under a Transaction Document and the Company does not timely perform its related obligations within the periods therein provided, then such Purchaser may rescind or withdraw, in its sole discretion from time to time upon written notice to the Company, any relevant notice, demand or election in whole or in part without prejudice to its future actions and rights; provided, however, that, in the case of a rescission of an exercise of a Warrant, the applicable Purchaser shall be required to return any shares of Common Stock subject to any such rescinded exercise notice concurrently with the return to such Purchaser of the aggregate exercise price paid to the Company for such shares and the restoration of such Purchaser’s right to acquire such shares pursuant to such Purchaser’s Warrant (including, issuance of a replacement warrant certificate evidencing such restored right).

5.14 Replacement of Securities. If any certificate or instrument evidencing any Securities is mutilated, lost, stolen or destroyed, the Company shall issue or cause to be issued in exchange and substitution for and upon cancellation thereof (in the case of mutilation), or in lieu of and substitution therefor, a new certificate or instrument, but only upon receipt of evidence reasonably satisfactory to the Company of such loss, theft or destruction. The applicant for a new certificate or instrument under such circumstances shall also pay any reasonable third-party costs (including customary indemnity) associated with the issuance of such replacement Securities.

5.15 Remedies. In addition to being entitled to exercise all rights provided herein or granted by law, including recovery of damages, each of the Purchasers and the Company will be entitled to specific performance under the Transaction Documents. The parties agree that monetary damages may not be adequate compensation for any loss incurred by reason of any breach of obligations contained in the Transaction Documents and hereby agree to waive and not to assert in any action for specific performance of any such obligation the defense that a remedy at law would be adequate.

5.16 Payment Set Aside. To the extent that the Company makes a payment or payments to any Purchaser pursuant to any Transaction Document or a Purchaser enforces or exercises its rights thereunder, and such payment or payments or the proceeds of such enforcement or exercise or any part thereof are subsequently invalidated, declared to be fraudulent or preferential, set aside, recovered from, disgorged by or are required to be refunded, repaid or otherwise restored to the Company, a trustee, receiver or any other Person under any law (including, without limitation, any bankruptcy law, state or federal law, common law or equitable cause of action), then to the extent of any such restoration the obligation or part thereof originally intended to be satisfied shall be revived and continued in full force and effect as if such payment had not been made or such enforcement or setoff had not occurred.

5.17 Independent Nature of Purchasers' Obligations and Rights. The obligations of each Purchaser under any Transaction Document are several and not joint with the obligations of any other Purchaser, and no Purchaser shall be responsible in any way for the performance or non-performance of the obligations of any other Purchaser under any Transaction Document. Nothing contained herein or in any other Transaction Document, and no action taken by any Purchaser pursuant hereto or thereto, shall be deemed to constitute the Purchasers as a partnership, an association, a joint venture or any other kind of entity, or create a presumption that the Purchasers are in any way acting in concert or as a group with respect to such obligations or the transactions contemplated by the Transaction Documents. Each Purchaser shall be entitled to independently protect and enforce its rights including, without limitation, the rights arising out of this Agreement or out of the other Transaction Documents, and it shall not be necessary for any other Purchaser to be joined as an additional party in any proceeding for such purpose. Each Purchaser has been represented by its own separate legal counsel in its review and negotiation of the Transaction Documents. For reasons of administrative convenience only, each Purchaser and its respective counsel have chosen to communicate with the Company through the Placement Agent Counsel, the legal counsel of the Placement Agent. Placement Agent Counsel does not represent any of the Purchasers and only represents the Placement Agent. The Company has elected to provide all Purchasers with the same terms and Transaction Documents for the convenience of the Company and not because it was required or requested to do so by any of the Purchasers. It is expressly understood and agreed that each provision contained in this Agreement and in each other Transaction Document is between the Company and a Purchaser, solely, and not between the Company and the Purchasers collectively and not between and among the Purchasers.

5.18 Saturdays, Sundays, Holidays, etc. If the last or appointed day for the taking of any action or the expiration of any right required or granted herein shall not be a Business Day, then such action may be taken or such right may be exercised on the next succeeding Business Day.

5.19 Liquidated Damages. The Company's obligations to pay any partial liquidated damages or other amounts owing under the Transaction Documents is a continuing obligation of the Company and shall not terminate until all unpaid partial liquidated damages and other amounts have been paid notwithstanding the fact that the instrument or security pursuant to which such partial liquidated damages or other amounts are due and payable shall have been canceled.

5.20 Construction. The parties agree that each of them and/or their respective counsel have reviewed and had an opportunity to revise the Transaction Documents and, therefore, the normal rule of construction to the effect that any ambiguities are to be resolved against the drafting party shall not be employed in the interpretation of the Transaction Documents or any amendments thereto. In addition, each and every reference to share prices and shares of Common Stock in any Transaction Document shall be subject to adjustment for reverse and forward stock splits, stock dividends, stock combinations and other similar transactions relating to shares of Common Stock that occur after the date of this Agreement.

5.21 WAIVER OF JURY TRIAL. IN ANY ACTION, SUIT, OR PROCEEDING IN ANY JURISDICTION BROUGHT BY ANY PARTY AGAINST ANY OTHER PARTY, THE PARTIES EACH KNOWINGLY AND INTENTIONALLY, TO THE GREATEST EXTENT PERMITTED BY APPLICABLE LAW, HEREBY ABSOLUTELY, UNCONDITIONALLY, IRREVOCABLY AND EXPRESSLY WAIVES FOREVER TRIAL BY JURY.

[Signature Pages Follow]

IN WITNESS WHEREOF, the parties hereto have caused this Securities Purchase Agreement to be duly executed by their respective authorized signatories as of the date first indicated above.

THERIVA BIOLOGICS, INC.

Address for Notice:

By: _____

Name:

Email:

Title:

Fax:

With a copy to (which shall not constitute notice):

[]

Attn:

Email:

*[REMAINDER OF PAGE INTENTIONALLY LEFT BLANK.
SIGNATURE PAGES FOR PURCHASERS FOLLOW.]*

[PURCHASER SIGNATURE PAGES TO SECURITIES PURCHASE AGREEMENT]

IN WITNESS WHEREOF, the undersigned have caused this Securities Purchase Agreement to be duly executed by their respective authorized signatories as of the date first indicated above.

Name of Purchaser:

Signature of Authorized Signatory of Purchaser:

Name of Authorized Signatory:

Title of Authorized Signatory:

Email Address of Authorized Signatory:

Facsimile Number of Authorized Signatory:

Address for Notice to Purchaser:

Address for Delivery of Warrant Shares to the Purchaser (if not same address for notice):

DWAC for Common Stock:

Subscription Amount: \$ _____

Shares of Common Stock: _____

Shares of Common Stock underlying the Pre-Funded Warrants: _____

Warrant Shares underlying the Common Warrants: _____

EIN Number: _____

☐ Notwithstanding anything contained in this Agreement to the contrary, by checking this box (i) the obligations of the above-signed to purchase the securities set forth in this Agreement to be purchased from the Company by the above-signed, and the obligations of the Company to sell such securities to the above-signed, shall be unconditional and all conditions to Closing shall be disregarded, (ii) the Closing shall occur on the second (2nd) Trading Day following the date of this Agreement and (iii) any condition to Closing contemplated by this Agreement (but prior to being disregarded by clause (i) above) that required delivery by the Company or the above-signed of any agreement, instrument, certificate or the like or purchase price (as applicable) shall no longer be a condition and shall instead be an unconditional obligation of the Company or the above-signed (as applicable) to deliver such agreement, instrument, certificate or the like or purchase price (as applicable) to such other party on the Closing Date.

[SIGNATURE PAGES CONTINUE]

Exhibit A
Form of Common Warrant
(See Attached)

Exhibit B
Form of Pre-Funded Warrant
(See Attached)

Exhibit C
Form of Lock-Up Agreement
(See Attached)
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